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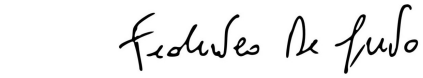
Scuola di Scienze Matematiche, Fisiche e Naturali

Corso di laurea triennale in fisica

Exploiting the Intrinsic Radioactivity of LYSO Crystals for the Measurement of the SiPM Breakdown Voltage for the BTL Detector

Candidate:
Giacomo Moretti - 896807

Supervisor:
Prof. Federico De Guio



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Introduction

The High-Luminosity Large Hadron Collider (HL-LHC) is a major upgrade to the existing Large Hadron Collider (LHC), one of the largest and most complex scientific instruments ever built. The primary objective of the HL-LHC, planned for operation in 2030, is to enhance and extend the physics reach of the LHC by significantly increasing both instantaneous and integrated luminosity. Specifically, the HL-LHC aims to increase the instantaneous luminosity by a factor of five beyond the original design value and the integrated luminosity by a factor of ten, targeting to deliver at least $250\,\mathrm{fb}^{-1}$ per year to each of the two high-luminosity, general-purpose detectors, ATLAS and CMS [1].

The MIP Timing Detector (MTD) is a new detector planned for the Compact Muon Solenoid (CMS) experiment during the HL-LHC era. The primary motivation for MTD arises from the anticipated increase in collision rates, which leads to a significant challenge known as pileup. With approximately 200 nearly-simultaneous pileup interactions expected in each bunch crossing, the existing CMS detector will require substantial upgrades to maintain its current performance.

The Barrel Timing Layer (BTL), the central section of the MTD detector extending up to $|\eta| = 1.48$, is designed to deliver a time resolution of the order of tens of picoseconds. The active elements of the BTL are thin bars of scintillating crystals. Each end of a crystal bar is coupled to a Silicon Photomultiplier (SiPM). The main challenges concerning the parameters of SiPMs for the BTL of CMS, which are crucial for its operation during the HL-LHC era, primarily relate to their radiation sensitivity and the resulting impact on the detector's time resolution. The two most critical parameters are the Dark Count Rate (DCR) and the Photon Detection Efficiency (PDE). Both PDE and DCR depend on the over-voltage (OV) applied to SiPM, and maintaining an optimal OV is essential to balance PDE and DCR and maximize time resolution [2].

Typically the SiPM's breakdown-voltage (BV) is extracted from its IV curve, but for BTL, once installed inside CMS, there will not be enough illumination to characterize the SiPMs with this approach. This work proposes a new procedure to

measure the BV of SiPMs exploiting the intrinsic radioactivity of the LYSO crystals coupled to each of them.

After a discussion of the physical framework and the challenges posed by the extreme data-taking conditions at the HL-LHC in Chapter 1, Chapter 2 prBVides a focus on SiPMs, LYSO:Ce crystals, an overview of the BTL layout and QA/QC procedure for the assembling process of each tray. Chapter 3 presents the current method used to characterize the over-voltage of the SiPMs and the new procedure developed using the behaviour of the energy peak in the spectrum of natural LYSO:Ce crystal radiation by performing an over-voltage scan using silicon photomultipliers.

Chapter 1

The HL-LHC Upgrade and The MTD Detector

The Large Hadron Collider (LHC) is a superconducting two-ring accelerator and collider, installed in the former 27 km-long tunnel of the LEP (Large Electron-Positron Collider). The LHC was designed to achieve very high collision energies, reaching up to 14 TeV in proton-proton collisions, and a maximum luminosity of 10^{34} $\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$. The main operational challenges of the LHC include managing beam–beam effects, minimizing beam emittance growth caused by phenomena such as intra-beam scattering or the electron cloud effect, maintaining magnetic field quality, and mitigating the impact of high radiation levels in critical zones of the accelerator [3].

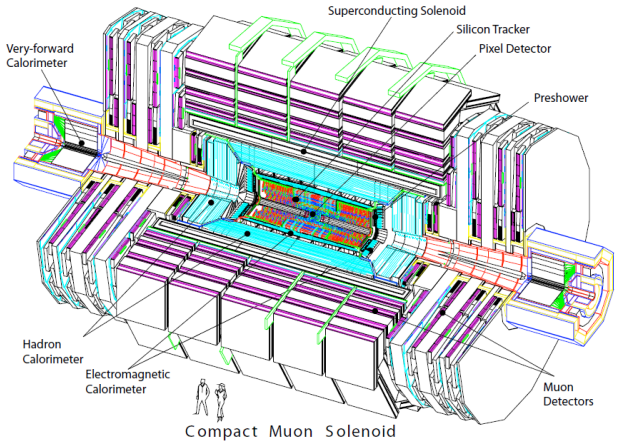


Figure 1.1: An cut version of the CMS detector [4].

The Compact Muon Solenoid (CMS) is a large-scale particle physics experiment and one of the two general-purpose detectors located at LHC (fig. 1.1). The detector is based on a large superconducting solenoid capable of generating a magnetic field of nearly 4 T. The primary goals of CMS are to explore physics at the TeV scale

and to study the mechanism of electroweak symmetry breaking, notably through the study of the Higgs boson. The discovery of the Higgs in Run 1 was enabled by several decay channels, in particular $H \to \gamma\gamma$ and $H \to ZZ^* \to 4\ell$, which provided clean signatures and precise mass measurements. In subsequent runs, additional channels such as $H \to WW^*$, $H \to \tau\tau$, and $H \to b\bar{b}$ have played a key role in probing its couplings and differential properties [4, 5].

1.1 The High Luminosity Phase

The High Luminosity LHC (HL-LHC) is the upgraded and enhanced phase of the Large Hadron Collider. The accelerator will enter soon in a shutdown period (Long Shutdown 3, LS3), and operations will resume in 2030, marking the beginning of the HL-LHC era, also referred to as Phase-2 of LHC operations.

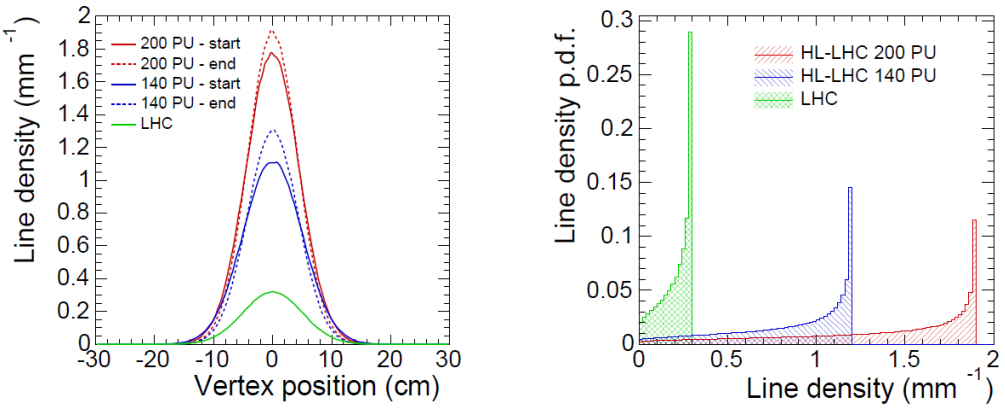


Figure 1.2: Left: Distribution of the vertices along the beam direction at the LHC (Run 1 and early Run 2) with approximately 30 pileup interactions, and at the HL-LHC with 140 and 200 pileup interactions. The solid (dashed) line refers to the start (end) of the fill. Right: Probability density functions of the line density along the beam axis for pileup of approximately 30, 140, and 200. The modes of the three distributions are 0.3, 1.2, and 1.9 mm^{-1}, and their means are 0.2, 0.9, and 1.4 mm^{-1}, respectively [1].

The main goal of the HL-LHC is to increase the luminosity, significantly enhancing the sensitivity for searches of physics beyond the Standard Model (BSM) and improving precision measurements of Standard Model processes, especially those involving the Higgs boson. To achieve this, the HL-LHC is designed to deliver a much higher instantaneous luminosity than its predecessor. Initially, it will be capable of reaching an instantaneous luminosity of 2×10^{35} $\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ at the start of

each fill. However, the nominal operational scenario foresees a leveled luminosity of 5.0×10^{34} cm^{-2} s^{-1}, maintaining an average of 140 interactions per bunch crossing (pileup) through continuous tuning of beam focusing and crossing profiles. An ultimate scenario envisions a leveled luminosity of 7.5×10^{34} cm^{-2} s^{-1}, resulting in about 200 pileup events per crossing.

This dramatic increase in luminosity presents major challenges for the current CMS detector, which will require substantial upgrades to continue operating efficiently. Key challenges include the significantly higher pileup, which can degrade the identification and reconstruction of the interactions of interest and increase false trigger rates (fig. 1.2). In addition, increased radiation damage that exceeds the tolerance levels of existing subdetectors and much higher data throughput demands require the upgraded detector to handle and transmit vastly greater data rates for storage and analysis [1, 2].

1.2 The MIP Timing Detector

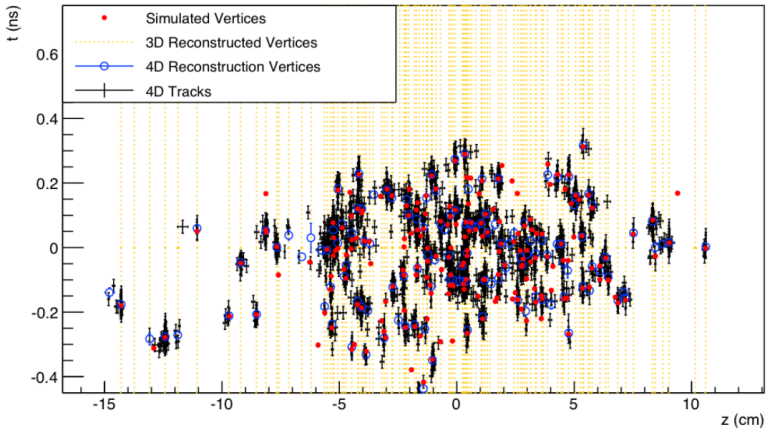


Figure 1.3: Simulated and reconstructed vertices in a bunch crossing with 200 pileup interactions assuming a MIP timing detector with 30 ps time resolution covering the barrel and endcaps. The horizontal axis is the z position along the beam line. The simulated vertices are the red dots. The vertical yellow lines indicate 3D-reconstructed (i.e. no use of timing information) vertices. The black crosses and the blue open circles represent tracks and vertices reconstructed using a method that includes the time information and is therefore referred to as "4D". Many of the vertices that appear to be merged in the spatial dimension are clearly separated when time information is available [2].

The MIP Timing Detector (MTD) is designed to introduce a completely new capability to CMS: a precise time measurement assigned to each Minimum Ionizing Particles (MIP) traversing it. This timing capability is instrumental in disentangling the numerous pileup interactions since individual interactions within a bunch crossing are distributed over time with an RMS of 180–200 ps. By associating charged tracks to their correct interaction vertices based on their timing information, the MTD greatly reduces the number of tracks from pileup vertices that are incorrectly associated with the hard-interaction vertex, effectively reducing the collision line density and recovering performance close to earlier LHC operations (fig. 1.3). The reduction of the number of tracks from pileup vertices that are incorrectly associated with the hard-interaction vertex is quantified in fig. 1.4. For a line density of 1.9 collisions per mm, which is the peak density for the case of 200 pileup collisions, the mean number of incorrectly associated tracks reaches over 20 without the use of timing information.

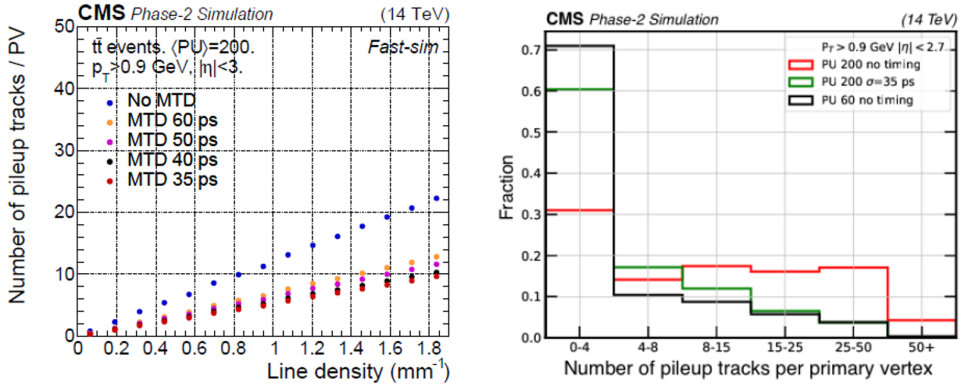


Figure 1.4: Left: Number of pileup tracks incorrectly associated with the hard interaction vertex as a function of the collision line density for different time resolutions. Right: Distribution of the number of incorrectly associated tracks with selection on timing information and without use of timing information. The vertical axis is the fraction of primary vertices which have the number of pileup tracks shown on the horizontal axis associated to them [2].

The detector is composed of two main sub-detectors to provide hermetic angular coverage up to a pseudorapidity η of ± 3 (fig. 1.5). The Barrel Timing Layer (BTL) is a cylindrical detector covering the barrel region up to $|\eta| = 1.48$ and is housed inside the Tracker Support Tube (TST). It utilizes LYSO:Ce crystals read out by Silicon Photomultipliers (SiPMs). The Endcap Timing Layer (ETL) uses Ultra-Fast Silicon Detectors (UFSDs), based on Low Gain Avalanche Detector (LGAD) technology sensors and consists of two disks instrumenting the endcap regions with $1.6 < |\eta| <$

3.0. Each disk features silicon devices on both sides to ensure hermetic coverage without gaps. The BTL is structurally integrated within the Tracker Support Tube (TST), between the Outer Tracker and the barrel ECAL, while the ETL is mounted on the endcap calorimeter nose cone, in the space between the Tracker and the Endcap Calorimeter.

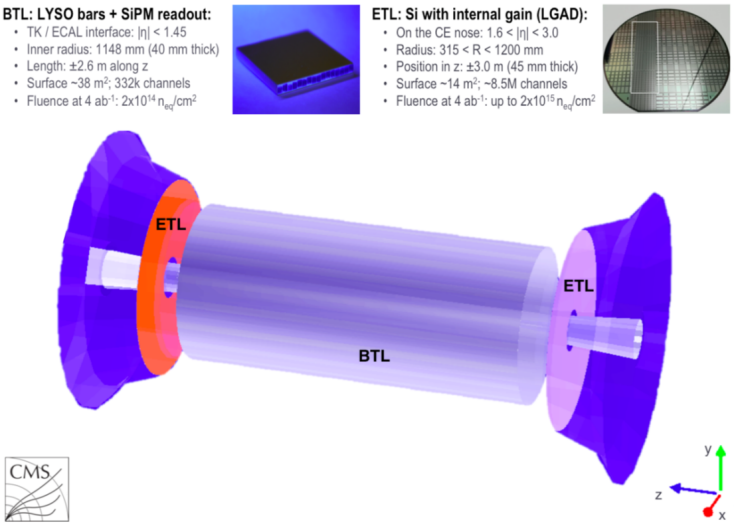


Figure 1.5: A schematic view of the geometry of the timing layers implemented for simulation studies, comprising a barrel layer (grey cylinder) at the interface between the tracker and the ECAL, and two silicon endcap timing layers (orange and light violet discs) located in front of the endcap calorimeter [2].

Beyond pileup mitigation, the MTD offers new capabilities such as charged hadron identification and searches for long-lived particles, improved reconstruction of electrons, muons, τ, photons, jets, and missing transverse momentum, enhanced precision in measurements of Standard Model processes — particularly in Higgs boson characterization and di-Higgs boson production searches — as well as unique opportunities in Heavy Ion physics through time-of-flight particle identification [2].

Chapter 2

The Barrel Timing Layer

The BTL is a thin cylindrical detector positioned inside the Tracker Support Tube, located between the outermost layer of the tracker and the beginning of the electromagnetic calorimeter barrel (ECAL). It extends between an inner radius of 1148 mm and an outer radius of 1188 mm, resulting in a maximum radial thickness of 40 mm. The detector covers an active length of approximately 5000 mm along the beam axis, with a total active surface area of about 38 m^2 (fig. 2.1) [2].

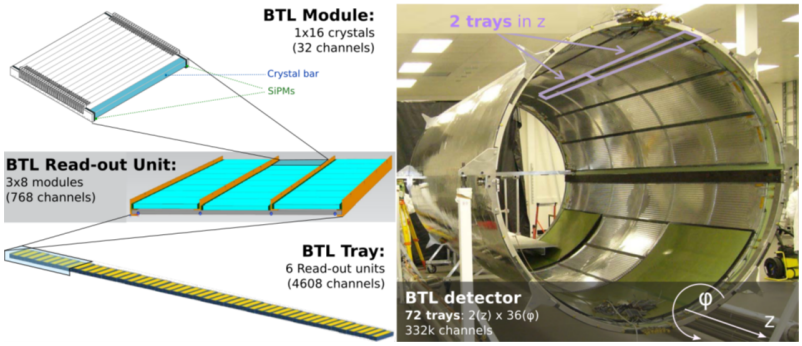


Figure 2.1: Overview of the BTL showing (left) the hierarchical arrangement of the various components, bars, modules, and Readout Units, and (right) trays (purple rectangles near the top) [2].

The sensitive elements of the BTL are thin bars of LYSO:Ce (Lutetium Yttrium Orthosilicate doped with Cerium) scintillating crystals. LYSO:Ce was selected due to its high light yield of around 40,000 photons per MeV, fast rise time below 100 ps, short decay time of about 40 ns, and high material density. LYSO:Ce is highly valued also due to its excellent resistance to radiation from gamma rays, neutrons, and charged hadrons [6]. Each end of a crystal bar is coupled to a Silicon Photomultiplier (SiPM), which is chosen for its compactness, robustness, insensitivity to magnetic fields, operation at low voltages, and tolerance to ambient light. The properties

and performance of both LYSO crystals and SiPMs will be discussed in detail in section 2.1.

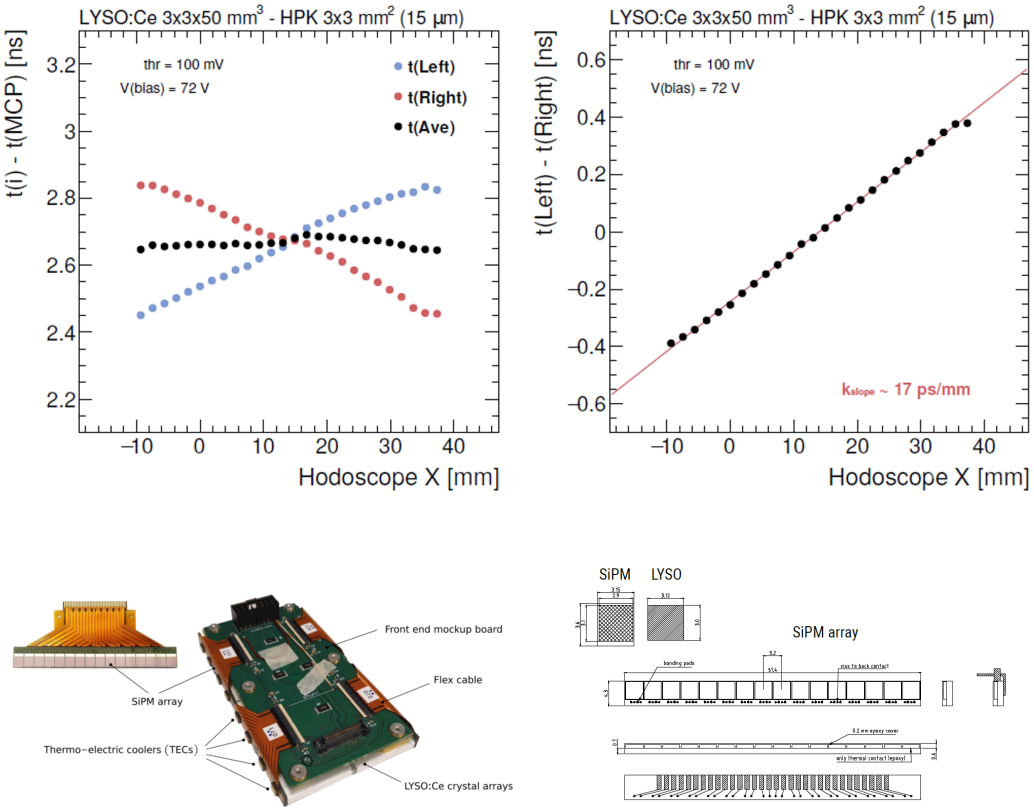


Figure 2.2: Left-top: time stamp from the left and right SiPMs and average time stamp, t_{ave}, as a function of the impact point X along the crystal bar axis. Right-top: difference between the two time stamps, t_{diff} [2]. Bottom: Pictures of a SiPM array (front side) and of a BTL detector module before insertion inside a copper housing, consisting of: 2 LYSO:Ce crystal arrays, 4 SiPM arrays including flex cables for signal routing to the front-end board, 4 Thermo-Electric Coolers (TECs), and one RTD temperature sensor on each array [7]. Drawings of a linear array of 16 SiPMs on a ceramic package showing the matching to the SiPM active area with the crystal dimensions [2].

The use of dual-ended readout—where each LYSO:Ce crystal bar is read out at both ends by SiPMs—offers one main advantage. It significantly improves the time resolution, as the timing information can be averaged between the two ends, reducing the impact of signal fluctuations and noise. Because light propagation in the crystal is uniform, the timing measurement is effectively taken as the average of the two ends, making it largely independent of the exact hit position as shown in fig. 2.2. This leads to more uniform and reliable time measurements across the

detector. For these reasons, the dual-ended configuration was chosen as the optimal solution for combining precise timing and robustness in the challenging environment of the HL-LHC.

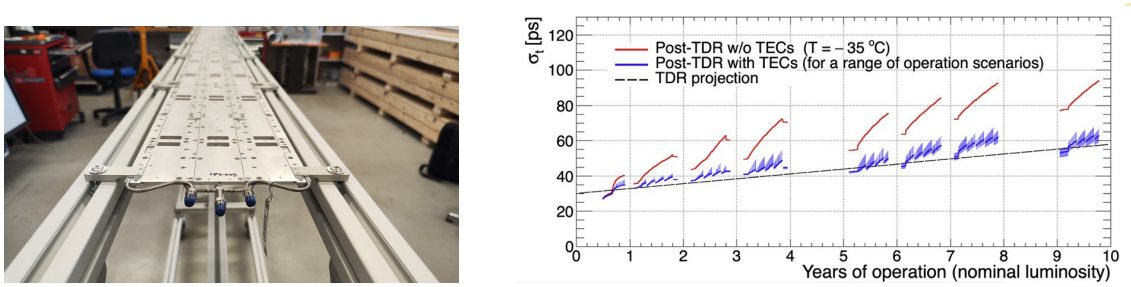


Figure 2.3: Left: CO_2 cooling pipes used to cool down the structural support, forming the cooling system of the BTL. Right: Improvement in the time detection performance due to annealing over time, despite the degradation of the SiPMs. A further enhancement is observed with the implementation of TECs (Thermoelectric Coolers) [8].

To cope with the expected radiation-induced damage, the BTL will operate at a temperature of about $-35\,°C$ using a two-phase CO_2 cooling system. In general, operating at low temperatures helps reduce the Dark Count Rate (DCR), thereby mitigating the effects of radiation damage and extend the lifetime of the SiPMs. In addition, periodic annealing cycles are foreseen to partially recover radiation-induced defects and maintain performance over time (fig. 2.3) [2].

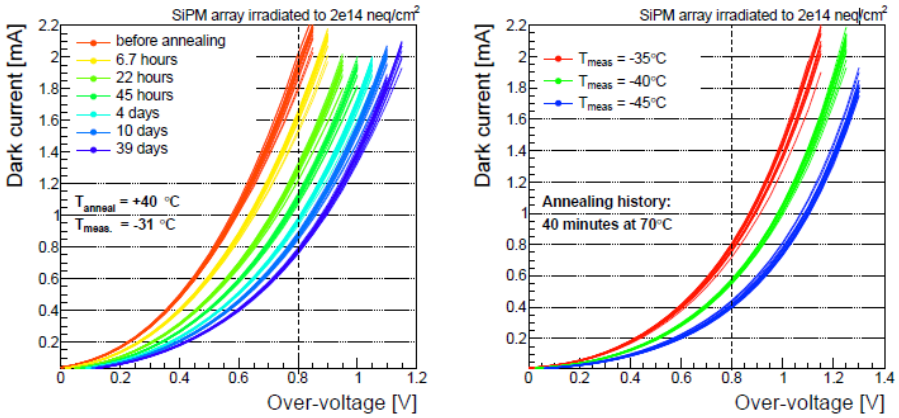


Figure 2.4: Left: Dark current as a function of over-voltage measured at $-31\,°C$ across all 16 channels of a SiPM array irradiated to 2×10^{14} 1 MeV n.eq./cm^2, after different periods of annealing at $40\,°C$. Right: Dark current of all 16 channels of a SiPM array irradiated to 2×10^{14} 1 MeV n.eq./cm^2 and annealed for 40 minutes at $70\,°C$, measured at different SiPM temperatures ranging from $-35\,°C$ to $-45\,°C$ [7].

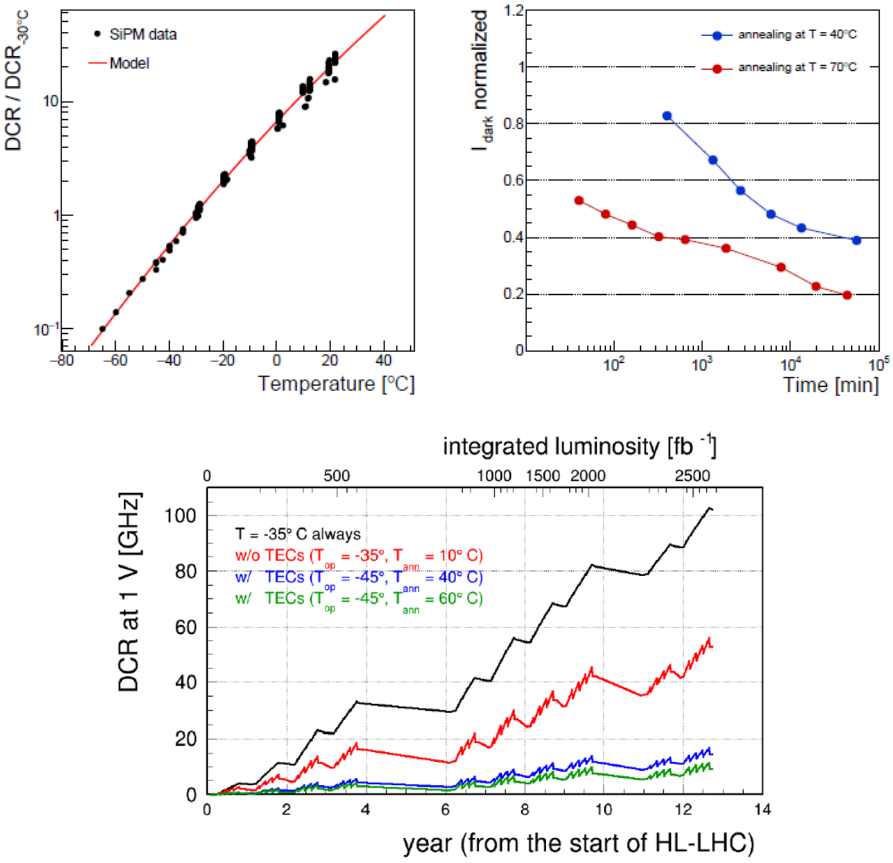


Figure 2.5: Top: Left – Dark count rate evaluated at 1 V over-voltage at a given temperature T, relative to the dark count rate at $T = -30\,°\text{C}$. Right – Reduction of dark current over time due to thermal annealing at 40 °C and 70 °C, normalized to the dark current measured after irradiation. Bottom: Evolution of dark count rate at fixed operating over-voltage of 1 V throughout the detector operation assuming the current baseline running plan of the HL-LHC and for different scenarios of SiPM temperature during operation and during technical stops when annealing can be accelerated [7].

The annealing is a recovery process observed in SiPMs after exposure to radiation damage, especially in high-energy physics experiments. Radiation creates defects in the silicon lattice of the SiPMs, leading to increased leakage current and consequently higher DCR. Annealing helps mitigate these adverse effects by partially restoring SiPM performance through a reduction in DCR (fig. 2.4). This is achieved by temporarily increasing the temperature of the device, so that the resulting thermal agitation allows some of the radiation-induced defects in the silicon lattice to reorganize or self-heal [9].

To address the deterioration of the detector's time resolution caused by radiation damage, Thermo-Electric Coolers (TECs), implemented using Peltier cells, were added to the backside of the SiPM package. One side of the TEC is thermally coupled to the SiPMs, while the other is connected to the Cooling Plate to dissipate the excess heat generated during the cooling process. The TECs can reduce the temperature of the SiPMs by an additional 10 °C with respect to the CO_2 temperature, bringing it down to −45 °C. Furthermore, they allow SiPM annealing at temperatures up to 60 °C [7].

The dark count rate (DCR) of SiPMs is strongly dependent on temperature, decreasing by approximately a factor of 1.9 for every 10 °C reduction. The impact of temperature on DCR, as well as the effectiveness of annealing strategies, is illustrated in the following figures in fig. 2.5.

2.1 The BTL sensors and layout

2.1.1 The LYSO:Ce Crystal

Lutetium Yttrium Oxyorthosilicate (LYSO) crystals are the scintillator materials most widely used today in PET detectors due to their convenient physical properties for the detection of 511 keV annihilation photons. Like SiPMs, they have become popular in high-energy physics. Doped with Ce^{4+}, this element plays an active role in the scintillation mechanism: excited electrons from the conduction band can be captured by Ce^{4+} ions, forming excited Ce^{3+} ions that de-excite by emitting light [10].

A particular feature of these crystals is their intrinsic radiation. Natural lutetium contains 2.6 % of ^{176}Lu, which undergoes β-decay to excited states of ^{176}Hf, producing a constant background signal (fig. 2.6). This intrinsic activity turned out to be useful for detector calibration, for assessing the overall performance, and for measuring the energy resolution at various energy levels [11]. In this work, the spectral shape resulting from the self-detection of both β-particles and γ-radiation has been

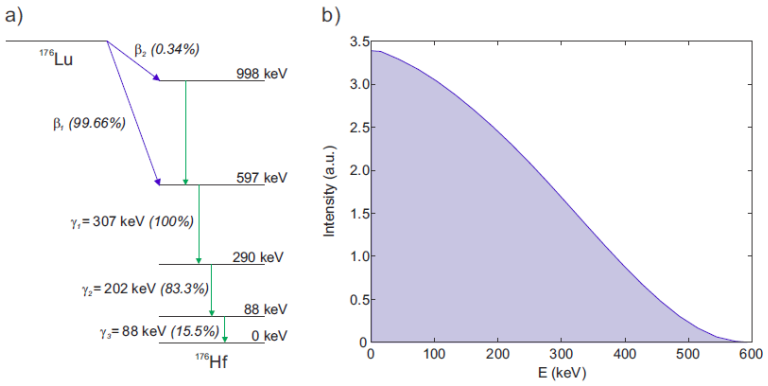


Figure 2.6: (a) Simplified ^{176}Lu decay scheme and (b) β-particle energy spectrum corresponding to the β_1 transition [11].

studied to determine the optimal parameter settings for SiPM calibration. As described later in section 2.1.3, all channels are read out by TOFHIR through the energy branches, where charge deposition is detectable. A parameter of attenuation has been introduced to attenuate the readout signal and shape the spectrum for optimal calibration [2, 12].

Each bar is 55 mm in length, 3.12 mm wide and 3.75 mm in thickness for best timing performance (fig. 2.7). Due to the position of the impact vertex, the radial thicknesses vary depending on the pseudorapidity to ensure uniform material traversal [13, 14].

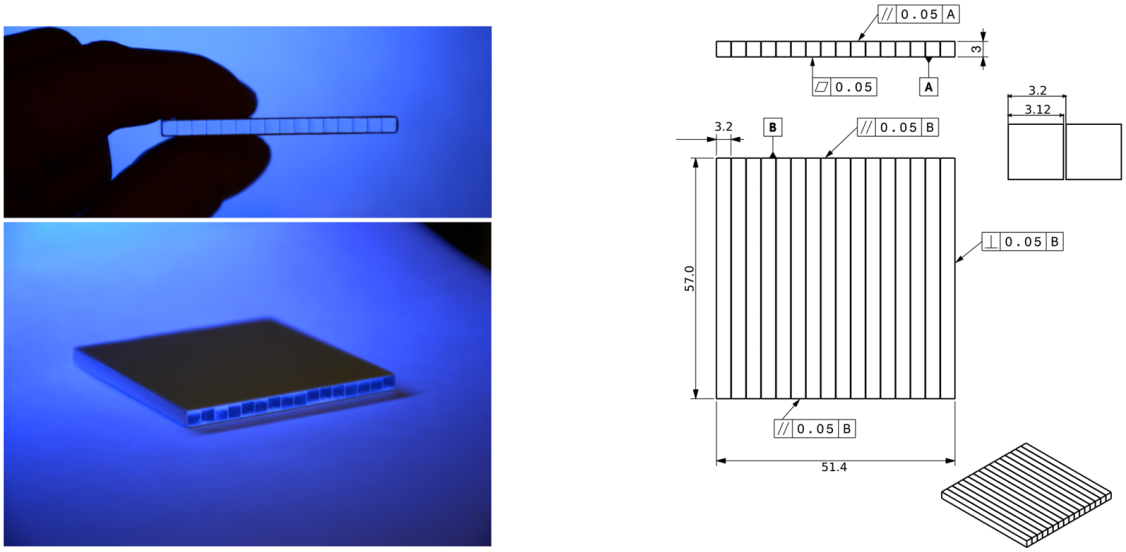


Figure 2.7: Left: Picture of a 16 × 1 array of crystal bars under UV illumination. Right: Drawing and dimensions of a BTL crystal array [2].

2.1.2 The Silicon Photomultiplier

The Silicon Photomultipliers, commonly known as SiPMs, are solid-state photodetectors that have gained considerable attention across various scientific and industrial fields. Their unique combination of high sensitivity, compactness, and excellent timing capabilities makes them ideal for applications ranging from single-photon detection to large-scale scintillation light measurements.

At the heart of an SiPM lies an array of tiny avalanche photodiodes operating in Geiger mode, known as Single-Photon Avalanche Diodes (SPADs). These microcells are capable of detecting single photons with high precision and amplifying the resulting electrical signal internally. This internal amplification, combined with their fast response times on the order of tens of picoseconds, provides SiPMs with remarkable temporal resolution. Moreover, SiPMs operate at relatively low voltages and are robust and compact, which further broadens their applicability.

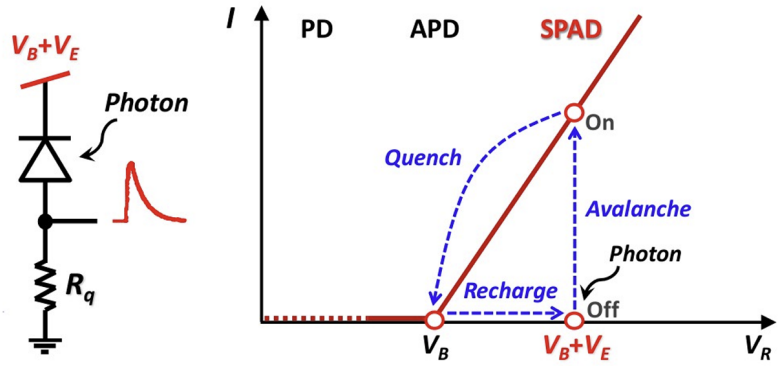


Figure 2.8: Schematic of a SPAD and its working mechanism. V_B represents the breakdown voltage; $V_E - V_B$ represents the over voltage; R_q is the quenching resistor, associated with a quenching voltage. The quenching system is used to drop the voltage and stop the avalanche, allowing the generation of a measurable pulse [15].

The fundamental building block of an SiPM, the SPAD, is essentially a diode reverse-biased above its breakdown voltage (VB or V_B). This reverse bias allows a single charge carrier generated by a photon interaction to trigger a self-sustaining avalanche, creating a measurable pulse (fig. 2.8). The amplitude of the pulse depends on the gain of the SiPM. The avalanche is then quenched by lowering the Bias Voltage (V_E), typically through a quenching resistor, preparing the cell for the next photon detection. The over-voltage (OV) is defined as the difference between the applied V_E and the breakdown voltage.

Understanding the intrinsic parameters of SiPMs is essential to harness their full potential. Characterizing a SiPM means measuring its Dark Count Rate (DCR),

which arises from thermally generated carriers and can be significantly reduced at low temperatures; other noise sources include afterpulsing and crosstalk These noise sources can degrade the detector's performance.

One key performance metric is the Photon Detection Efficiency (PDE), which depends on the quantum efficiency, avalanche triggering probability, and the effective fill factor of the device.

Temperature significantly influences SiPM performance. Variations in temperature cause shifts in the breakdown voltage, which in turn affect the gain (fig. 2.9, right), while the Dark Count Rate typically decreases as temperature lowers (fig. 2.9, left). Although some parameters remain stable if the over-voltage is kept constant, practical systems commonly incorporate temperature compensation mechanisms [9].

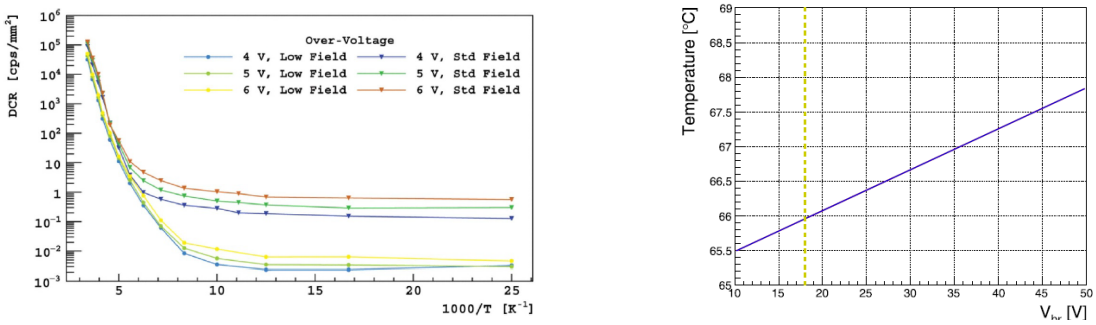


Figure 2.9: Left: DCR for NUV-HD-SF (Standard Field) and NUV-HD-LF (Low Field) devices as a function of 1/T and over-voltage [9]. Right: Increase of breakdown voltage with temperature for SiPM BT80. The dashed yellow line indicates the breakdown voltage measured at room temperature (18 °C) [16].

The successful deployment of SiPMs in high-energy physics experiments depends on a synergistic approach that combines radiation-hardened power regulation, sub-20 ps timing resolution, and efficient thermal management [17, 16]. The high noise levels after irradiation complicate the measurement of key parameters such as gain, PDE, and breakdown voltage.SiPMs remain operational after intense irradiation; combining cryogenic operation and thermal annealing can largely restore their performance, but with a gradual loss during the life-time of the HL-LHC [18, 19, 20, 21]. In BTL, a time resolution of about 30 ps at the beginning of operation and about 60 ps at the end of operation are achieved with 3.75 mm thick crystal arrays coupled to 25 µm cell size SiPMs produced by Hamamatsu Photonics (HPK) [14].

2.1.3 Detector layout

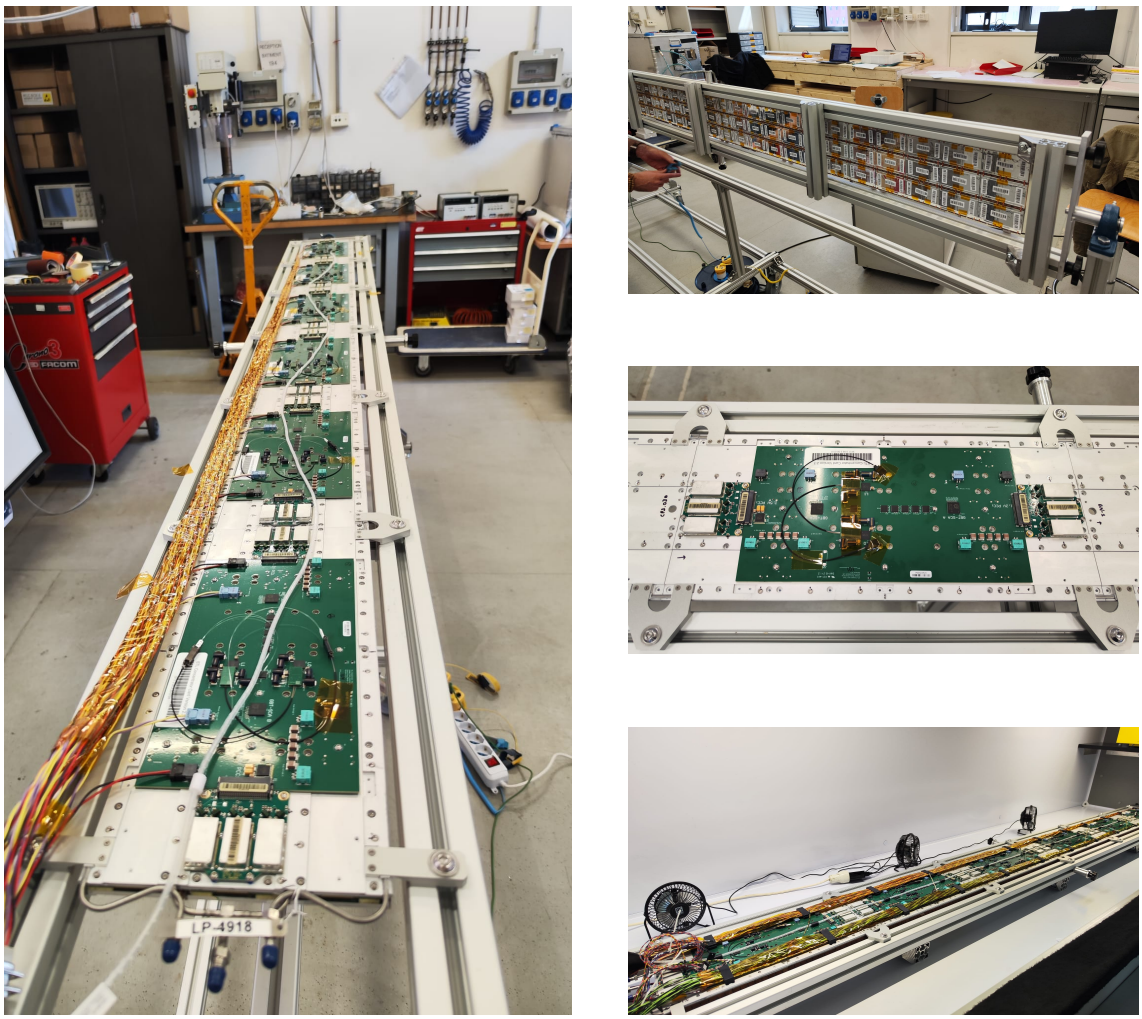


Figure 2.10: Left: Cabling of the first tray assembled in Milan. All six readout units are connected with optical fibers (white) and sensing cables (orange). The cooling pipes are also visible. Top-right: All detector modules (DMs) mounted on the bottom of the tray. Center-right: The first CC board assembled. Bottom-right: The first completed tray is shown inside the cooling chamber, which is equipped with ventilation.

The BTL is subdivided into 72 identical segments called trays. Each tray consists of a three-layer structure (fig. 2.11), and can be further divided into six sections referred to as Readout Units (RUs). Each RU, like the tray itself, is structured in three layers: a Sensor Layer, comprising LYSO:Ce crystal bars optically coupled to SiPMs, assembled into sensor modules and connected to the front-end electronics; a Cooling Plate providing structural support, thermal dissipation, and the mounting points for sensor modules and the data concentration electronics. The readout

ASIC, called TOFHIR, is integrated within the sensor layer, housed together with the LYSO crystals and SiPMs inside the copper structure. On the opposite side of the cold plate, the data readout boards—comprising the Concentrator Cards (CC) and the Power Converter Cards (PCC)—are connected through openings in the cold plate.

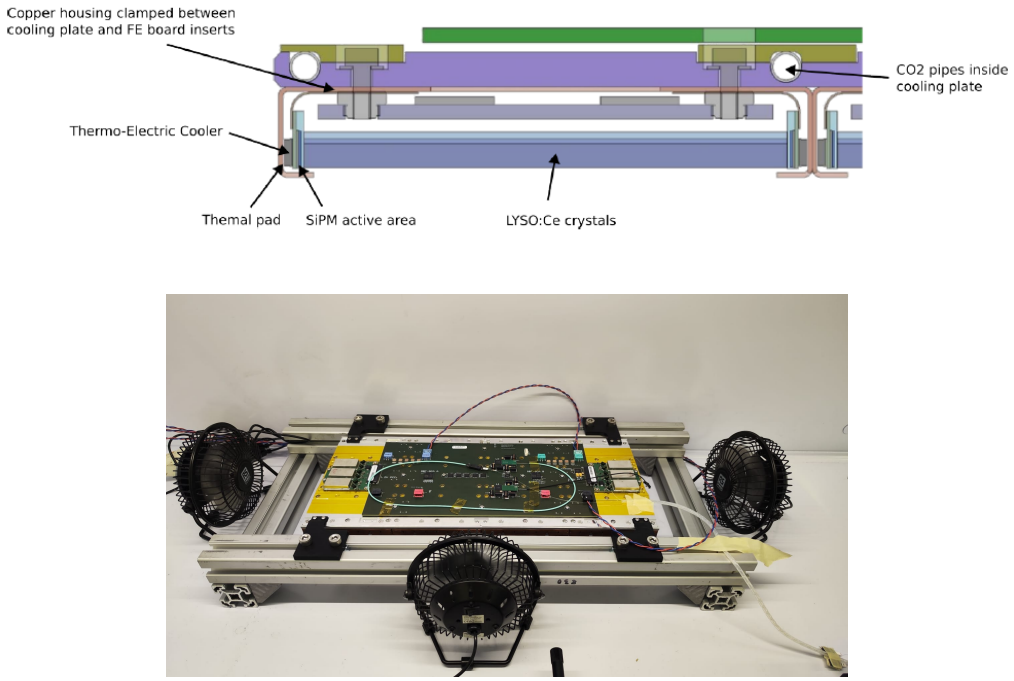


Figure 2.11: Top: Transverse section of a BTL detector module mounted on the cooling plate. The green bar on top represents the Concentrator Card (CC) shown in the image below [7]. Bottom: Photograph of a complete Readout Unit (RU), illustrating the three-layer structure of the BTL tray: the Sensor Layer (located beneath the structural frame, consisting of 768 SiPMs and 384 LYSO crystals); the Cooling Plate, which also provides structural support; and the readout section, which includes the Concentrator Cards. The RU is read out via optical fiber (with wire) and is mounted onto a mechanical support using black brackets.

The TOFHIR2 ASIC, specifically designed for time-of-flight measurements at high event rates, provides a Time-to-Digital Converter (TDC) resolution of 12 ps. The SiPM output is replicated across multiple branches, including one dedicated to energy measurement—based on integrated charge and mapped one-to-one to 1024 Analog-to-Digital Converter (ADC) channels—and another dedicated to timing detection. The ASIC features a test pulse, which is a signal generated for testing, characterization, and calibration purposes. Additionally, TOFHIR2 incorporates radiation tolerance features and is specifically optimized for the fast scintillation

signals produced by LYSO crystals and detected by SiPMs [12, 22]. Data are transmitted to the Concentrator Card, which also manages configuration, synchronization, reset, and trigger signals. Power Converter Cards provide regulated power to the Readout Units.

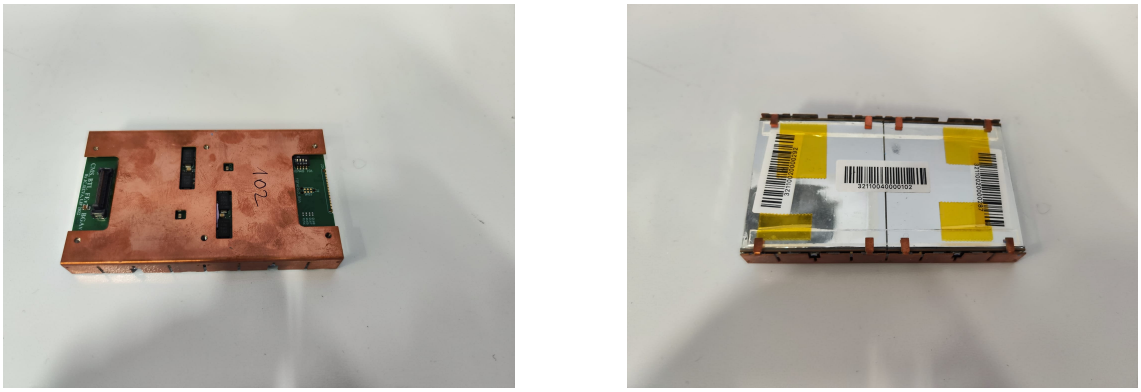


Figure 2.12: A detector module. On the right, the two individual sensor modules can be observed.

A single sensor module is composed of Lutetium Yttrium Orthosilicate crystals doped with Cerium (LYSO:Ce). Each crystal bar is read out by a pair of Silicon Photomultipliers (SiPMs). Associated with each sensor module is a TOFHIR2 ASIC. Two sensor modules form a Detector Module (DM) (fig. 2.12).

The modularity per tray includes 6 Readout Units (RUs), 24 Front-End boards (4 per RU), 6 Concentrator Cards (one per RU), 144 sensor modules (24 per RU), 72 detector module (12 per RU) 4608 readout channels (SiPMs) per tray (32 SiPMs per sensor module), and 2304 crystal bars (16 per sensor module) [2].

2.2 BTL Assembly and Quality Control

A rigorous Quality Assurance and Quality Control (QA/QC) process is essential in the assembly and validation of detector systems to ensure reliable performance during operation. In particular, it is crucial to verify that all readout channels are fully functional and that each detector module has been correctly assembled, especially when involving critical steps such as optical coupling or glue application. The primary goal of QA/QC is to identify any defective components or assembly issues before final integration. This includes confirming that every channel responds correctly to input signals, and that all glued modules are optically active and capable of detecting and transmitting light as expected. One of the main risks in the absence of proper QA/QC procedures is that some modules, while appearing mechanically in-

tact, may in fact be non-functional due to imperfect optical coupling, misalignment, or air bubbles in the glue that prevent light transmission. These undetected issues can compromise the performance of the entire system once installed, and may be extremely difficult or impossible to correct afterward. For this reason, QA/QC testing is a necessary step to validate the functionality and uniformity of all components, ensuring that the system meets its design specifications before deployment.

The detector is currently being constructed through a coordinated effort across four international assembly centers: two located in the United States, one in China, and one at the University of Milano-Bicocca in Italy. As part of this effort, the Milano-Bicocca center developed three dedicated tests to be run on the fully assembled tray with the goal of validating and monitoring the quality of the local assembly process: a study of the behaviour of energy peaks in the spectrum of natural LYSO:Ce crystal radiation; a test pulse to verify the TDC resolution of 20 ps; and the activation of the TECs to verify the temperature variation in the SiPMs. All tests were conducted inside a cold storage chamber maintained at a constant temperature of 18 °C.

I focused on the accurate identification of the LYSO scintillation peak, developing and implementing a robust fitting procedure to ensure stable and reliable peak extraction. This step was essential for verifying the energy response of the system and for establishing a consistent calibration method. In addition, I carried out a temperature stability study to monitor and understand the thermal behaviour of the system during the test cycles, an aspect that is crucial for the performance and calibration of the SiPM-based readout. This study was aimed at ensuring the absence of shifts in the response of individual SiPMs due to temperature gradients across the module, which could otherwise affect the accuracy of the measurements. As part of this work, I actively participated in the initial testing campaigns of the first detector trays carried out in Milan. This hands-on involvement was a key opportunity to contribute directly to the commissioning of the system. These early activities were fundamental in validating the test procedures and refining the data analysis tools used throughout the QA/QC phase.

The qualification tests performed on the readout system provide essential metrics to assess the functionality, uniformity, and overall performance of each component—ranging from full Detector Modules (DMs) and Sensor Modules (SMs) down to individual readout channels. Based on the outcomes of the three main tests described, a set of acceptance and rejection criteria had been defined to identify and flag potential issues.

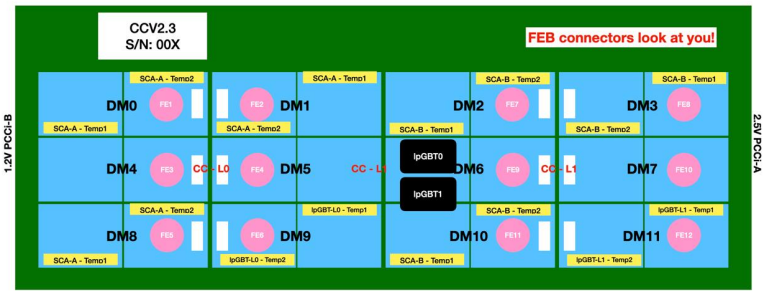


Figure 2.13: A schematic of a readout unit. The yellow boxes indicate the available temperature sensors.

2.2.1 Uniformity test of readout channels

In the first test, using double-sided detection from the SiPMs placed at each end of a single bar, a complete radiation spectrum was obtained by averaging the two histograms collected from both ends of the crystal, as shown in fig. 2.14-top. The goal of this test is to verify the response uniformity of the readout channels, ensuring consistent signal detection across both ends of the crystal and identifying any discrepancies that could indicate channel imbalance or defects. These measurements are made possible by the energy branch of the TOFHIR ASICs, which enables the readout of all ADC channels, as described in section 2.1. Once a more detailed plot was obtained, software was used to detect the peak corresponding to the γ energy of 300 keV in the spectrum. The key observable is the position and shape of the gamma peak across different channels. To be accepted, the peak position in each channel should fall within a specified tolerance window. The peak width (e.g., FWHM or sigma) should be consistent with the nominal energy resolution, and the responses from both ends of the same crystal bar should be symmetric. The channel-to-channel variation observed in fig. 2.14-bottom arises mainly from intrinsic differences in the SiPM sensors and the associated readout electronics. These include small fluctuations in SiPM gain and noise, variations in the ADC channels and front-end ASIC response, as well as non-uniform optical coupling between the LYSO crystal and each SiPM. Such variations are expected in a complex multichannel system and must be characterized and calibrated to ensure uniform signal detection. Significant deviations beyond typical tolerances could indicate malfunctioning channels or defects requiring further investigation. Channels showing no peak, highly distorted spectra, strong asymmetry, or unexpected gain behaviour (such as pedestal shifts) are rejected, as they indicate issues like poor coupling or malfunctioning components.

Both the attenuation setting and the SiPM over-voltage (OV) were tuned to exploit at best the ADC dynamic range.

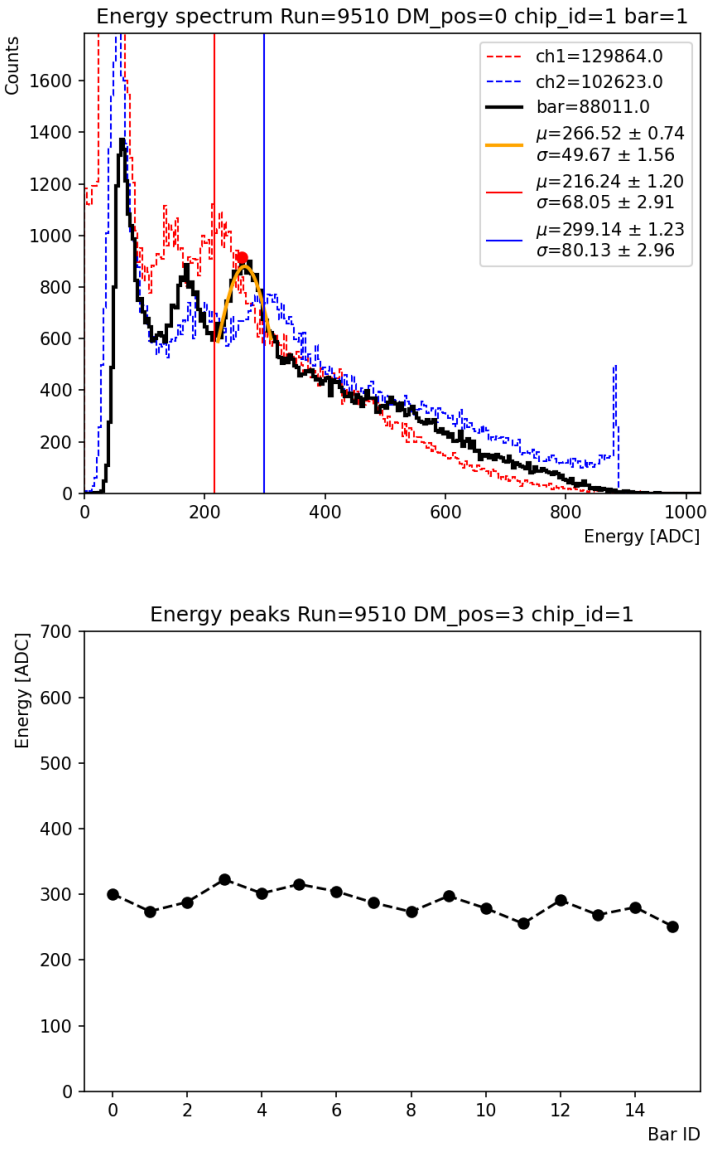


Figure 2.14: Top: The red and blue spectra show, respectively, channel one and channel two of a single bar. A channel corresponds to the signal from one side of the bar, read by a single photomultiplier. The black spectrum represents the average of the two channels. Bottom: All the ADC values of the peaks are shown with respect to the bars in a single sensor module.

2.2.2 Thermal stability test with biased Thermo-Electric Coolers

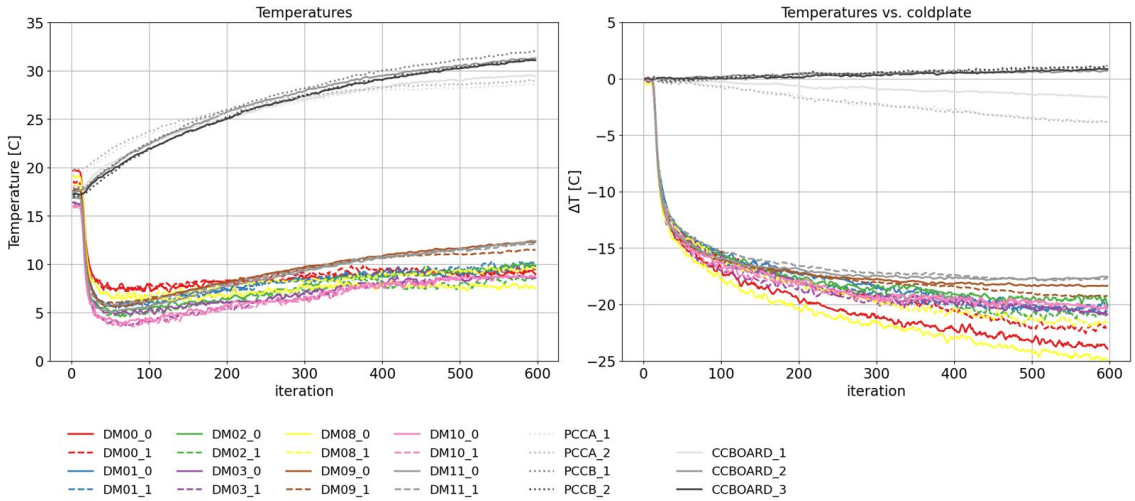


Figure 2.15: Left: Temperature of a readout unit during a TEC test. The cooling of the detector modules (DMs) and the heating of the CC board can be observed. The test was conducted without the cooling system, which will later provide heat mitigation for the setup. Right: Temperature difference over multiple iterations.

In the second test (fig. 2.15), the Thermo-Electric Coolers (TECs) were biased while monitoring the temperature difference between the SiPM package and the Cooling Plate. This measurement allowed verification of the TECs' capability to effectively reduce the SiPM temperature and maintain thermal stability, which is essential for ensuring optimal detector performance. Two plots are inspected during this test: one showing the temperature decrease of a readout unit over time (iterations), as recorded by the sensors, and the other displaying the temperature difference measured by each sensor relative to the cooling plate. The non-uniformity observed in the temperature readings of individual sensors is primarily attributed to residual miscalibrations of the ADC channels used for their measurement, but also to the differences between the TECs. While such discrepancies can be minimized through careful calibration, they are rarely eliminated entirely. All temperature sensors should exhibit coherent behaviour, with variations within the expected calibration uncertainty. Rejection criteria include failure of the TECs to achieve the required ΔT , incoherent or inconsistent thermometer readings, or the presence of instabilities such as thermal oscillations.

During the test, a temperature difference (ΔT) of approximately 20 °C between the SiPM package and the cooling plate was achieved. While this demonstrates

effective cooling performance under controlled conditions, achieving similar thermal stability at the operational target temperature of around −45 °C presents additional challenges.

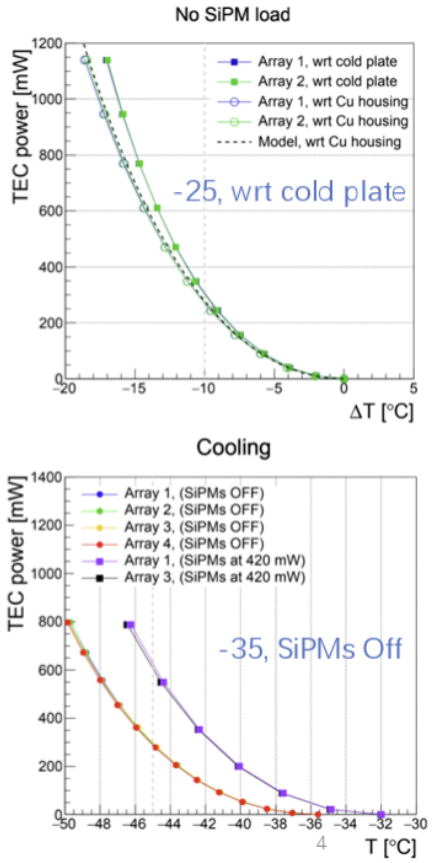


Figure 2.16: Top: TEC power as a function of ΔT with respect to the elements listed (two cold plates and two copper housings), starting at −25 °C, compared with a plot showing the modelled behaviour. Bottom: TEC power versus temperature with various curves for different SiPMs (two turned on, four turned off), all starting at −35 °C [23].

Cold tests of the DM modules were conducted at PKU using a specific module inside a constant temperature chamber equipped with cold plates (fig. 2.16)[23], confirming the ability to reach temperatures of 45 °C. In terms of thermal cycles, a total of 100 thermal cycles were also performed on two types of DMs verifying the long-term stability of the thermal contact of the DMs[24].

Cold tests on a complete tray were successfully conducted at CERN to demonstrate realistic and safe operation with CO_2 at −35 °C and SiPMs at −45 °C. TEC tests at each temperature identified optimal power settings to reach SiPM targets, with corresponding ΔT values recorded. The system demonstrated stable perfor-

mance at −45 °C in the Tray Integrated Facility (TIF) environment [25].

2.2.3 Time resolution characterization with digital Test Pulse

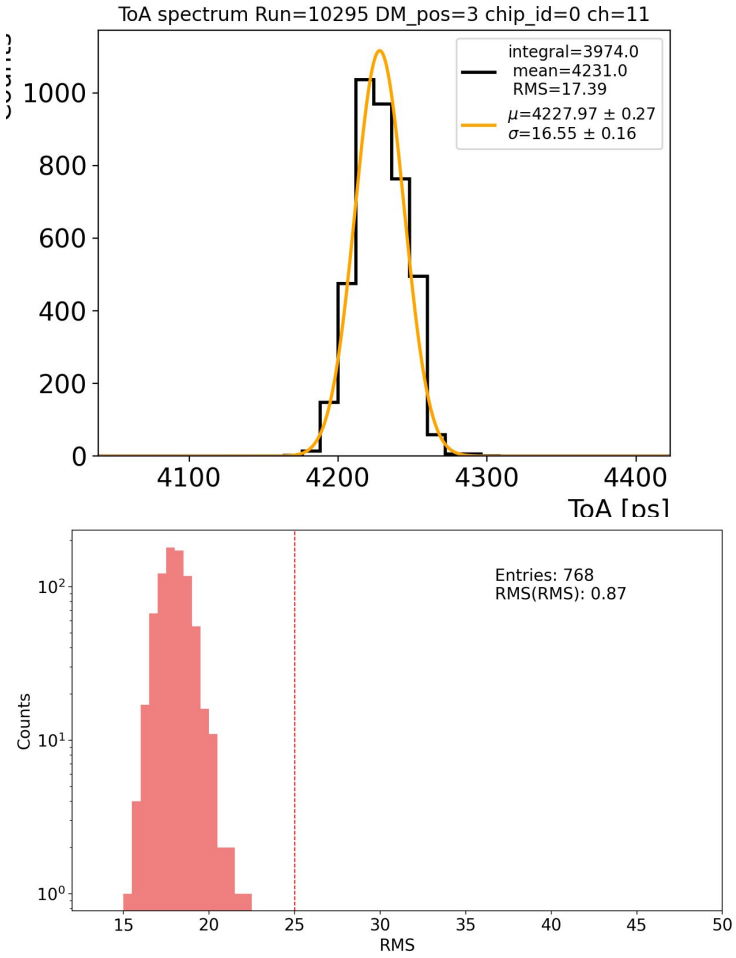


Figure 2.17: Top: Time difference measurement for a single channel. Bottom: Summary plot showing the time differences for all channels within a readout unit. The key metric used to evaluate the system performance is the RMS (or sigma) of the distribution, which quantifies the time resolution.

In the third test (fig. 2.17), the timing resolution of the digital electronics chain was evaluated using a digital test pulse. Time differences between multiple channels were measured, and the resulting distribution was analysed. The standard deviation of this distribution provided a direct estimate of the system's intrinsic time resolution, which was found to be on the order of 20 ps.

This test also verified that all readout channels respond correctly and synchronously, ensuring consistent time measurements across the system. Because the

trigger signal is purely digital—free from the fluctuations and jitter typical of physical detector signals—this method allows for a precise and reproducible evaluation of the front-end electronics' intrinsic timing performance. Specifically, the test measures the relative time alignment between channels when stimulated simultaneously, a critical feature for applications requiring accurate time-of-arrival information, such as time-of-flight measurements and fast event reconstruction. The key metric is the RMS (sigma) of the distribution of time differences between channels under simultaneous stimulation.

The timing resolution of an individual channel, σ_{single}, is a fundamental figure of merit for the acquisition system's performance. Assuming two channels have identical and independent timing resolutions, the standard deviation of the time difference $\Delta t = t_{\text{left}} - t_{\text{right}}$ between the two channels is related to the single-channel resolution by $\sigma_{\Delta t} = \sqrt{2}\,\sigma_{\text{single}}$.

Using the purely digital test pulse, the distribution of time differences Δt was measured across channels. The observed standard deviation $\sigma_{\Delta t}$ was approximately 17 ps, leading to an estimated single-channel resolution of

$$\sigma_{\text{single}} = \frac{\sigma_{\Delta t}}{\sqrt{2}} \approx \frac{17\,\text{ps}}{\sqrt{2}} \approx 12\,\text{ps}.$$

This result aligns well with the expected performance from the TDC design specifications. All channels must satisfy this timing resolution criterion to be accepted.

For the Barrel Timing Layer (BTL) system, however, the primary metric used to evaluate timing performance is the resolution on the mean time, defined as $t_{\text{mean}} = (t_{\text{left}} + t_{\text{right}})/2$. Assuming equal timing resolutions for the two channels, the uncertainty on the mean time is given by

$$\sigma_{\text{mean}} = \frac{\sigma_{\text{single}}}{\sqrt{2}} = \frac{\sigma_{\Delta t}}{2}.$$

As a complementary approach, and one that is actively employed in test beam campaigns at CERN, the timing resolution can also be evaluated using real detector signals. By selecting events occurring near the center of a scintillator bar—where the signal propagation times to both ends are approximately equal—arrival times t_{left} and t_{right} are recorded.

The distributions of t_{mean} and Δt are then analyzed, with their standard deviations σ_{mean} and $\sigma_{\Delta t}$ providing independent and robust estimates of the system's timing performance. Consistency between σ_{mean} and $\sigma_{\Delta t}$ validates the timing calibration. According to BTL design expectations, the target timing resolution on the mean time, including all contributions, is approximately 30 ps.

Chapter 3

Qualification of the SiPMs for BTL

Considering the sixteen crystal bars of a single sensor module and the SiPM array formed by all SiPMs on the same end of the bars, the current response of the system was analyzed as a function of the applied over-voltage to the array. The SiPMs were uniformly illuminated using an external light source to ensure a consistent response across all channels. The purpose of this measurement is to accurately extract the breakdown voltage VB of each SiPM in the array. This allows to verify the uniformity of the electrical behaviour across all channels and to ensure that the detectors operate within the expected specifications. Specifically, the aim is to identify possible deviations or anomalies that may compromise performance, such as unexpected shifts in VB due to assembly defects, handling, or environmental conditions. Accurate knowledge of this parameter is essential for setting the appropriate operating voltage and guaranteeing consistent gain and timing characteristics across the module.

To extract the breakdown voltage, the current–voltage (IV) characteristic curve was measured for each array, and the point of maximum of the derivative $\mathrm{d}(\log I)/\mathrm{d}V$ was identified. This method, widely adopted for SiPM characterization, provides a reliable and reproducible estimate of the breakdown voltage, corresponding to the inflection point in the logarithmic response curve.

It is important to note, however, that the breakdown voltage determined through this method does not correspond exactly to the physical onset of avalanche multiplication in the SiPM microcells. Since the derivative is computed from a current that is already increasing, the device is, in practice, already active when the maximum is observed. Nevertheless, this approach provides a well-defined reference point, and the breakdown voltage can be consistently redefined up to a global offset without

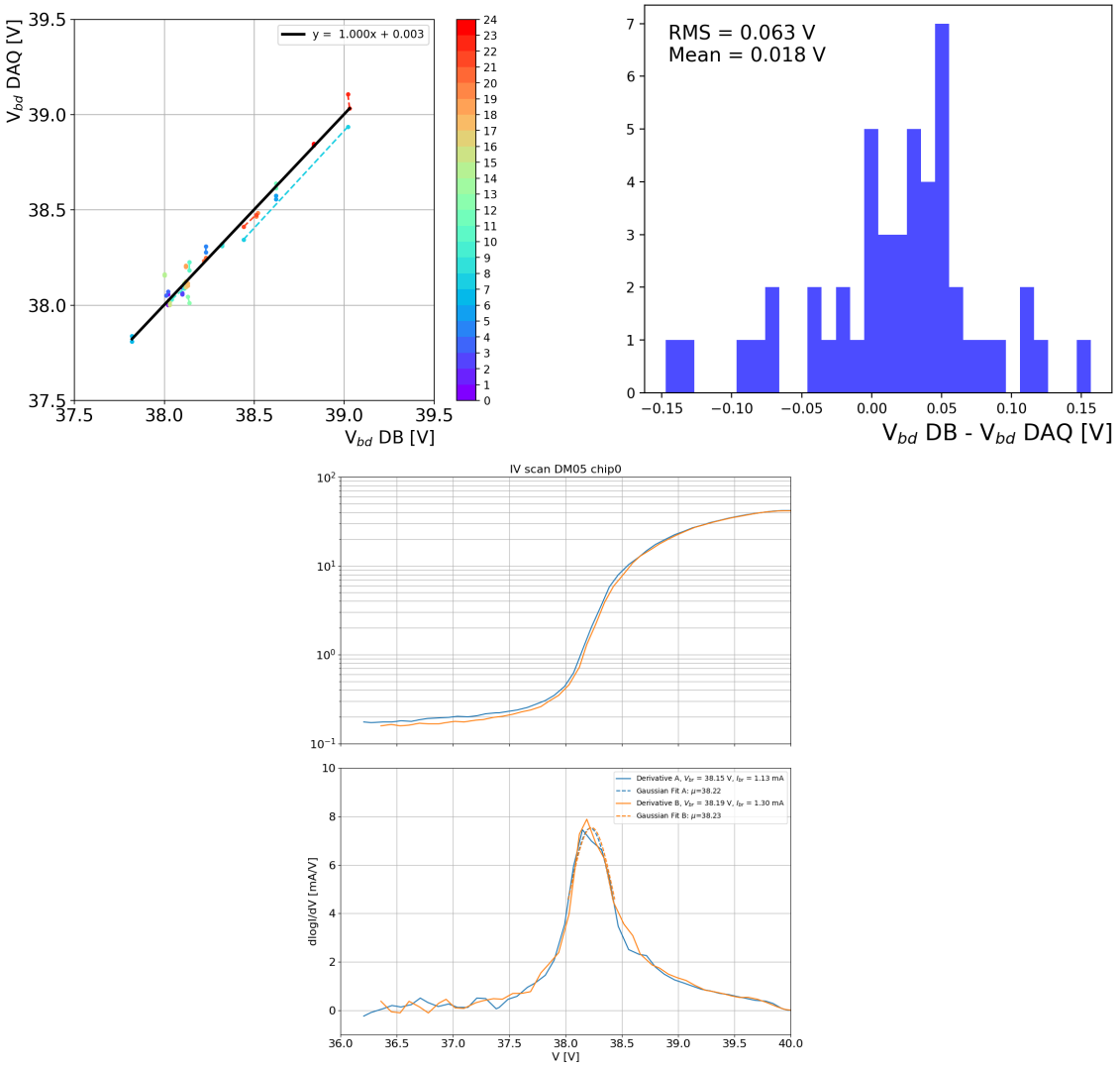


Figure 3.1: Top-left: Correlation map of breakdown voltage (VB) estimation. DAQ values correspond to the measured breakdown voltages, while DB values are those provided by the manufacturer (HPK). Top-right: Distribution of the difference between the expected (manufacturer) and the measured (VB) values. Bottom: IV scan curve consisting of two plots: the upper plot shows the logarithmic scale of $\log I$ versus voltage for two SiPM arrays (indicated by the orange and blue lines), and the lower plot presents the derivative $\mathrm{d}(\log I)/\mathrm{d}V$ with a Gaussian fit used to identify the peak corresponding to the breakdown voltage. The two colored lines represent the two SiPM arrays within a single sensor module.

affecting the validity of the characterization. As such, it remains a conventional but effective method to align measurements and define the operating over-voltage ($\mathrm{OV} = V_{\mathrm{bias}} - VB$).

Using the extracted breakdown voltages, a correlation map was generated to compare the measured values against the values provided by the SiPM HPK. This comparison allows for the identification of potential deviations due to handling, assembly, or environmental factors, and serves as an additional quality control check to ensure that all channels operate within specifications. All of this is illustrated in fig. 3.1, which shows both the IV curves with their logarithmic derivatives and the correlation map with the values from the producer.

To achieve sufficient accuracy with this method, it is necessary to inject light into the SiPM—particularly when measuring small SiPMs or devices with low gain and low dark count rate which is the case. One advantage of the IV method is that it is straightforward to perform, and in general, these characteristic measurements are fast and well-suited for device monitoring during the production process [9]. However, a major drawback is the requirement for an external light source, which will not be available as soon as the BTL detector is installed inside CMS. This limitation motivated the development of a new method that exploits the crystal's intrinsic radioactivity to determine the SiPM's VB. The details of this approach are presented in Sec. 3.2.

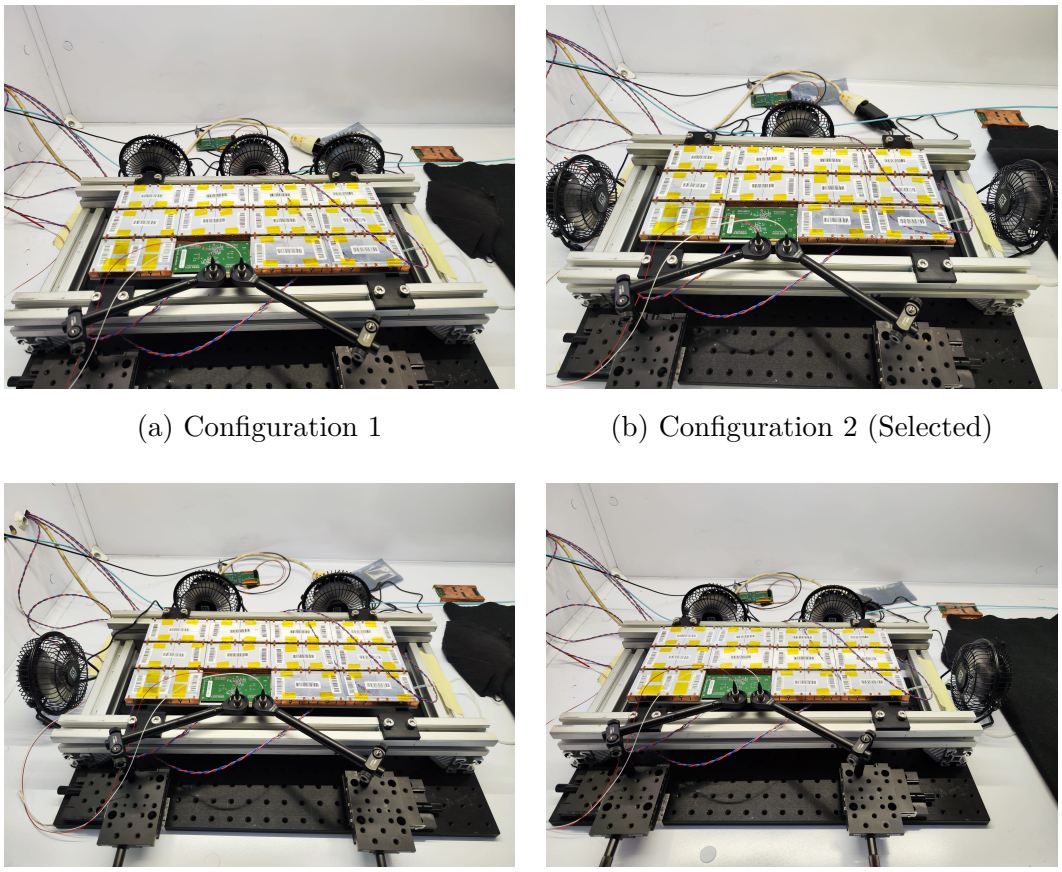
3.1 Temperature dependence

Due to the strong temperature dependence of SiPM parameters — such as the breakdown voltage, which shifts by approximately $37\,\mathrm{mV}\,^{\circ}\mathrm{C}^{-1}$ — a preliminary study was conducted using a single Readout Unit (RU). The aim of this study was to identify the optimal configuration to ensure thermal stability during operation. For this reason, the RU was placed inside a temperature-controlled environment (a refrigerator), and several fan configurations were tested to promote uniform airflow around the setup, as shown in fig. 3.2. Each configuration was evaluated based on its ability to minimize local temperature variations and maintain overall thermal stability.

The final temperature from each twenty-one sensors was recorded six times, with each measurement taken after a complete acquisition cycle performed by executing an IV scan. Between cycles, the fan configuration was changed to test all possible setups, including one without ventilation.

Fig. 3.3 shows the final temperatures measured by each Detector Module (DM) during the qualification runs (horizontal axes). Each run corresponds to a complete

acquisition cycle under a specific fan configuration. Multiple runs were performed for each of the four configurations, resulting in a range of temperature values per run due to differences among DM sensors. The vertical axes represents the final temperatures of individual DMs, allowing assessment of both absolute temperature levels and uniformity across modules for each cooling setup. While fans significantly reduced the overall temperature compared to no ventilation, differences between all the configurations were minimal, indicating stable thermal conditions regardless of the setup.



(a) Configuration 1

(b) Configuration 2 (Selected)

(c) Configuration 3

(d) Configuration 4

Figure 3.2: Matrix view of the four fan configurations evaluated for thermal management. Configuration 2 (top-right) was selected for all measurements due to its optimal thermal stability and compatibility with the setup.

For each fan configuration, the temperature data from all sensors were averaged over the six runs to obtain a representative temperature profile. The variation across the RU was then quantified by calculating the standard deviation (STD) of the average temperatures measured on each DM.

The optimal configuration was selected by minimizing this temperature spread,

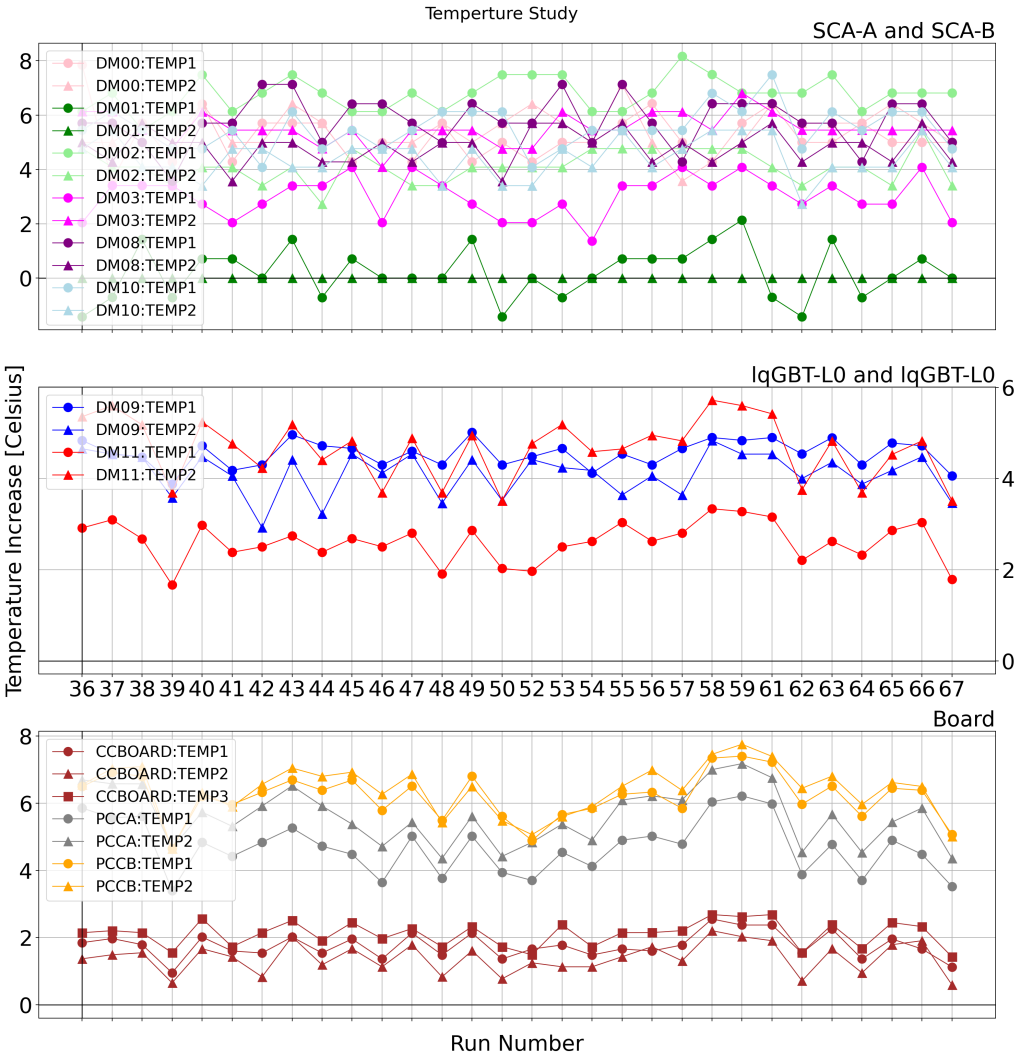


Figure 3.3: Final temperature recorded by each Detector Module (DM) during the qualification runs. Each point corresponds to the temperature measured by a single DM in a given run. Multiple runs were performed under four different fan configurations, including no ventilation. The plot highlights minimal temperature variation between setups, confirming stable thermal behaviour throughout the acquisition period.

achieving the best uniformity across the RU.

Configuration 2 — with fans positioned at the right, left, and top sides of the RU in fig. 3.2 — showed the lowest STD of 0.88 °C, compared to values ranging from 0.96 °C to 1.13 °C for the others. Despite small overall variations, this setup provided the most stable thermal conditions and was therefore used for subsequent measurements.

It is important to note that the STD includes not only true temperature differences but also systematic uncertainties from sensor calibration and ADC offsets. Nonetheless, the low spread confirms effective thermal uniformity across the RU.

During the Uniformity Test of the Readout Channel, described in section 2.2, the temperature was monitored using the same sensors used before placed throughout the RU. This test aimed to ensure and verify temperature uniformity across the entire RU. As shown in fig. 3.4, the temperature increases with time (or readout cycles), as expected during data acquisition. However, the key observation is that the temperature remains remarkably uniform across the RU, indicating effective thermal distribution.

From the graph, it is also evident that the Power Concentrator Cards (PCCs) experience roughly twice the temperature increase compared to the Detector Modules (DMs). This is expected, as the PCCs provide power and thus dissipate more heat. Conversely, the Control Card (CC) shows about half the temperature increase of the DMs, consistent with its role as purely readout electronics with lower power consumption.

Although sensors from DM9 and DM11 are plotted separately due to the different granularity of their ADC readout, it is noticeable that despite greater fluctuations in their measurements, these sensors reach similar temperature deltas as the other DMs. This confirms the overall temperature uniformity across all sensors.

This uniformity suggests that the cooling system, composed of fans, plays a crucial role in maintaining consistent thermal conditions during operation.

In conclusion, an optimal fan configuration was identified that ensures the best thermal uniformity across the Readout Unit (RU). This was confirmed both during the IV scan measurements and the Uniformity Test of the Readout Channel, which will be used going forward to perform the measurements by directly comparing results with those from the IV scans.

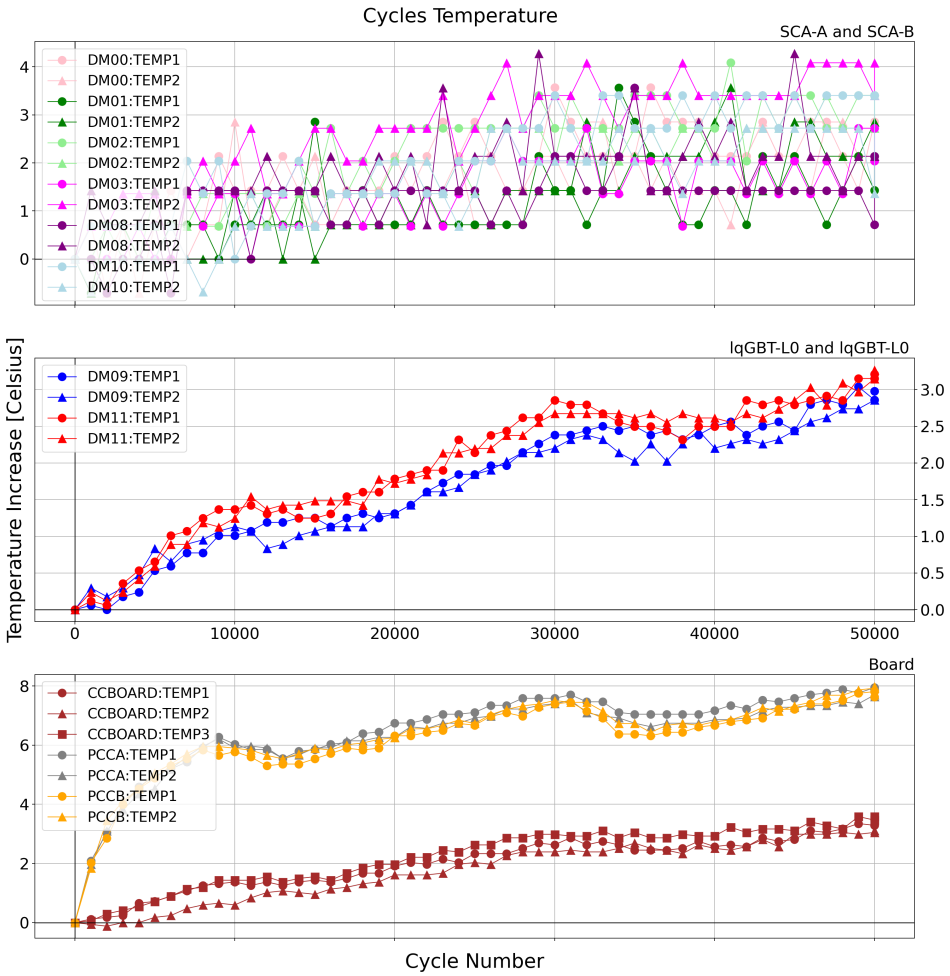


Figure 3.4: Temperature monitoring as a function of the number of readout cycles of the TOFHIR (50k cycles in 15 minutes). As shown, the sensors on the CC board are displayed separately from those on the detector modules (DMs). Additionally, sensors from DM9 and DM11 are reported separately as they are read out by an ADC with different granularity. The graph clearly shows the temperature increase trends across different sensor groups, highlighting uniformity despite variations in readout granularity.

3.2 Exploiting the LYSO intrinsic radioactivity to measure the SiPM VB

Considering the behaviour of the energy peak in the spectrum of natural LYSO:Ce crystal radiation during an over-voltage (OV) scan with a SiPM, the reconstructed energy extracted from the ADC signal, E_{ADC}, follows the relation:

$$E_{\mathrm{ADC}}(OV) \propto E_{\mathrm{dep}} \times \mathrm{Gain}(OV - VB_{\mathrm{shift}}) \times \mathrm{PDE}(OV - VB_{\mathrm{shift}}) \tag{3.1}$$

where:

- E_{ADC} denotes the reconstructed energy, obtained from the position of the signal peak in ADC units,
- E_{dep} is the energy deposited in the crystal by the interacting radiation,
- *Gain* is the SiPM gain, which increases approximately linearly with the corrected over-voltage,
- PDE is the photon detection efficiency, which also increases with the corrected over-voltage, though with a weaker dependence,
- VB_{shift} is a free parameter introduced to correct the applied over-voltage and estimate the breakdown voltage.

As the applied over-voltage increases, the position of the energy peak in ADC units shifts accordingly, driven by the combined effect of the increasing *Gain* and PDE (fig. 3.5). This dependence can be exploited to characterize the SiPM response and monitor the system's stability during operation.

The *Gain* and PDE curves used in the fit are not theoretical models but experimentally determined functions. Specifically, they are obtained by performing an IV scan on each SiPM, estimating the breakdown voltage VB as the maximum of $\mathrm{d}(\log I)/\mathrm{d}V$, and then scanning the response as a function of over-voltage $OV = V_{\mathrm{bias}} - VB$.

The same IV-based breakdown voltages were used for all SiPMs, and a free parameter V_{shift} was introduced in the fitting function. This parameter accounts for any discrepancy between the breakdown voltage estimated from the IV measurement and that obtained using the new method.

The breakdown voltage VB of each SiPM was estimated by analysing the 307 KeV energy peak in the spectrum of the Lutetium acquired from both ends of a single bar while scanning the over-voltage. The data were collected using the dedicated

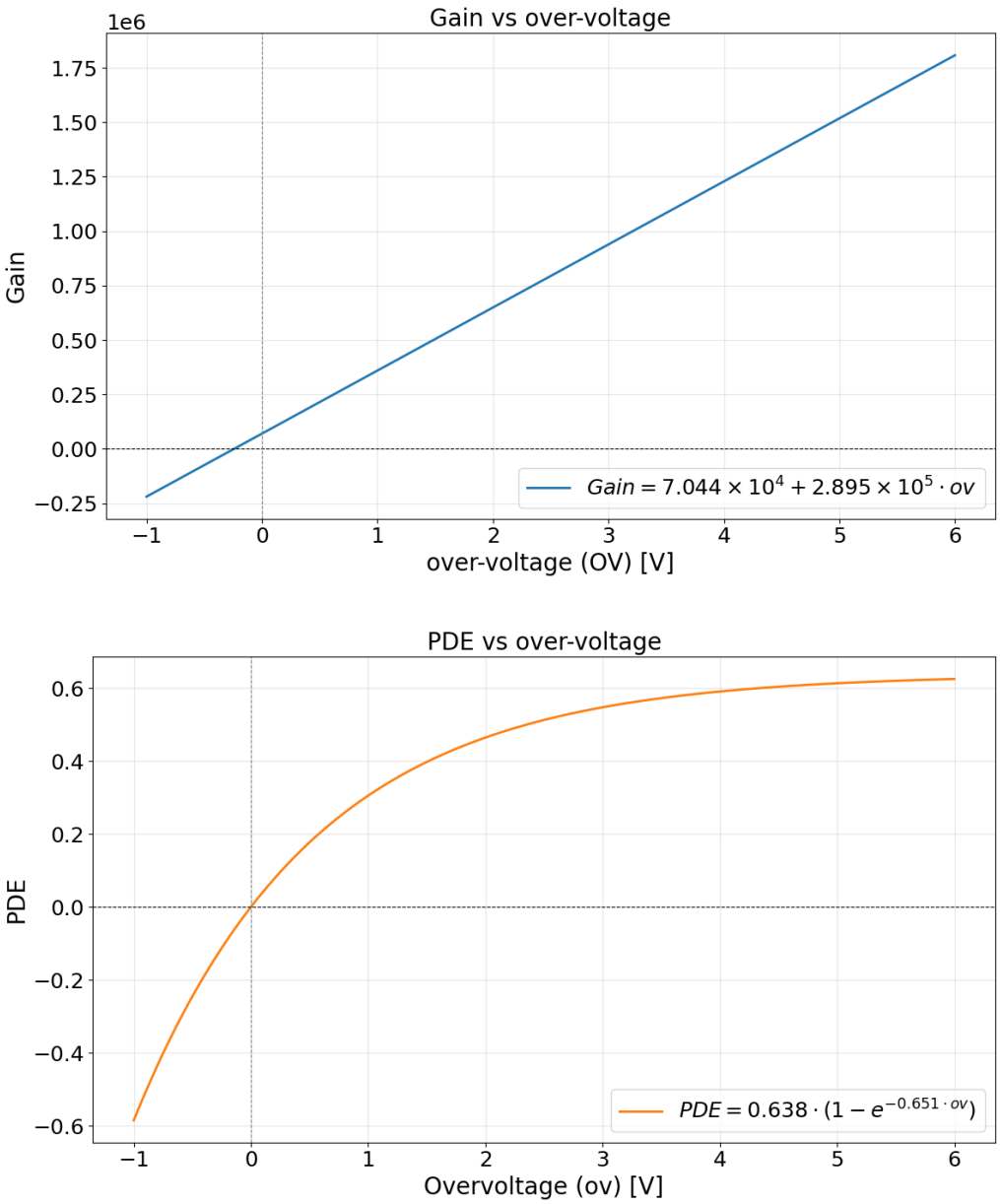


Figure 3.5: Top: Behavior of the *Gain* as a function of over-voltage for different SiPMs. Bottom: Behavior of the PDE as a function of over-voltage for different SiPMs. Both parameters contribute to the increase in ADC peak position as described in eq. (3.1).

software developed for the Uniformity Test of the Readout Channels, as described in section 2.2.

A breakdown voltage shift VB_{shift} was extracted for each SiPM by fitting the measured response with the model defined in eq. (3.1). For each array side, a single value of VB_{shift} was then obtained by averaging the shifts measured across all sixteen bars.

This approach provides a consistent estimation and comparison to the breakdown voltages originally inferred from IV-curve analysis, ensuring full coherence between the functional model used in the fit and the actual SiPM's behaviour observed under operation.

The development of this new method was carried out entirely on a single Readout Unit (RU), which was not mounted on a tray. This RU was specifically built for testing purposes and served as a workbench. Ultimately, the estimated breakdown voltages of the SiPM arrays were applied to one of the six RUs on an assembled tray.

3.3 Data analysis and interpretation

The need to identify the peak corresponding to the 307 keV γ energy directly from the SiPMs prompted the development of an alternative method for its extraction.

Once the peak was roughly identified, the new method—well developed and fully functional for detecting the peak in the non-averaged spectrum—ultimately provides an estimate of the peak position within an energy interval expected to be found in the individual spectra at each end of the crystal bar (see fig. 3.6, top). In the figure, the blue and red vertical lines mark the positions of the two peaks detected in the spectrum, representing signals originating from the opposite ends of the crystal bar.

An estimate of the uncertainty on the position of the Lu peak was obtained from the analysis of 37 runs acquired under identical conditions. The peak positions from each run are reported in the histogram shown in fig. 3.6, bottom. This analysis confirms that the reproducibility of the measurement is within approximately 2%. This value was used as the uncertainty on the data points.

This uncertainty estimation was performed only for a single over-voltage (OV) value — specifically, the one selected in the uniformity test of readout channels described in section 2.2. The estimated error was then applied to all other OV values used in the study. For the remaining OV values, the ADC peak was measured only once.

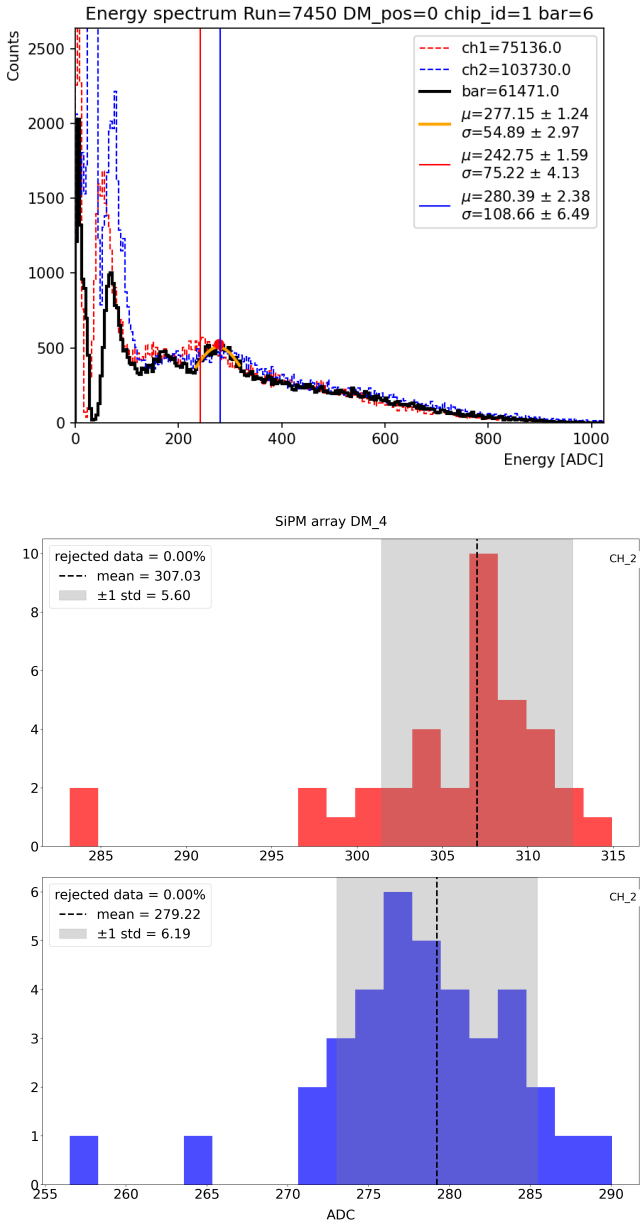


Figure 3.6: Top: Example of a single non-averaged spectrum from an individual SiPM channel. Although the peak is not clearly visible due to noise and resolution, the method enables identification of the peaks corresponding to signals from both ends of the crystal bar. The blue and red vertical lines indicate the positions of the detected peaks, corresponding respectively to the signals from the two ends of the crystal bar. Bottom: Distribution of the peak positions from single runs at fixed over-voltage detected in a single bar of one of the two sensor modules within a detector module. CH indicates the two channels corresponding to signals from the two SiPMs, one on each side of the crystal bar. The percentage of rejected data is used to estimate the number of successfully detected peaks.

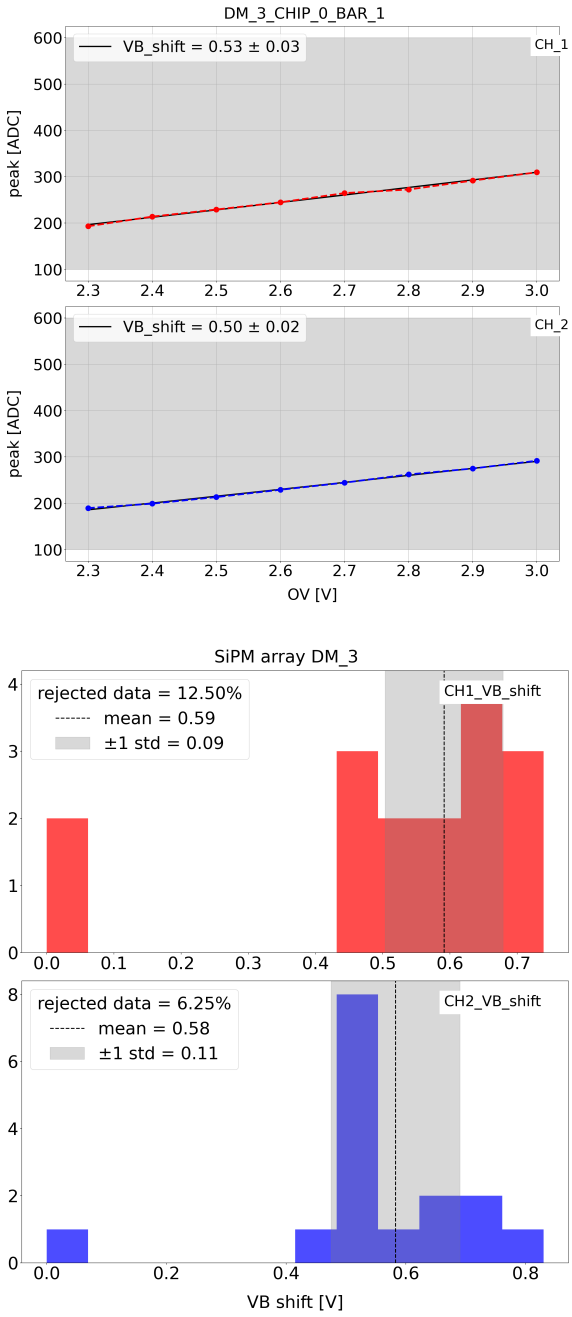


Figure 3.7: Top: Fit of the energy peak positions as a function of over-voltage for the two channels (left and right SiPM of a bar) of a single LYSO bar. The gray band indicates the accepted voltage range used for the linear fit. Bottom: Histograms of the breakdown voltage values extracted from the 16 individual SiPMs composing each SiPM array (one per channel). These distributions are used to calculate the mean and standard deviation, providing an estimate of the breakdown voltage for each array.

In addition, for each SiPM, the occurrence and distribution of values rejected due to failed peak detection were monitored and incorporated into the overall assessment of data quality and reliability.

An accurate fit was performed to reliably extract the breakdown voltage shift, as illustrated in fig. 3.7.

The TOFHIR involves multiple adjustable parameters, one of which is the attenuation setting. I carefully selected the attenuation value to optimize the dynamic range of the ADC, ensuring that all relevant peaks fall within the energy window under consideration. This choice is crucial to maximize the resolution and accuracy of the measurements. As a selection criterion, I used the percentage of failed fits and undetected peaks, minimizing it to ensure optimal peak reconstruction efficiency.

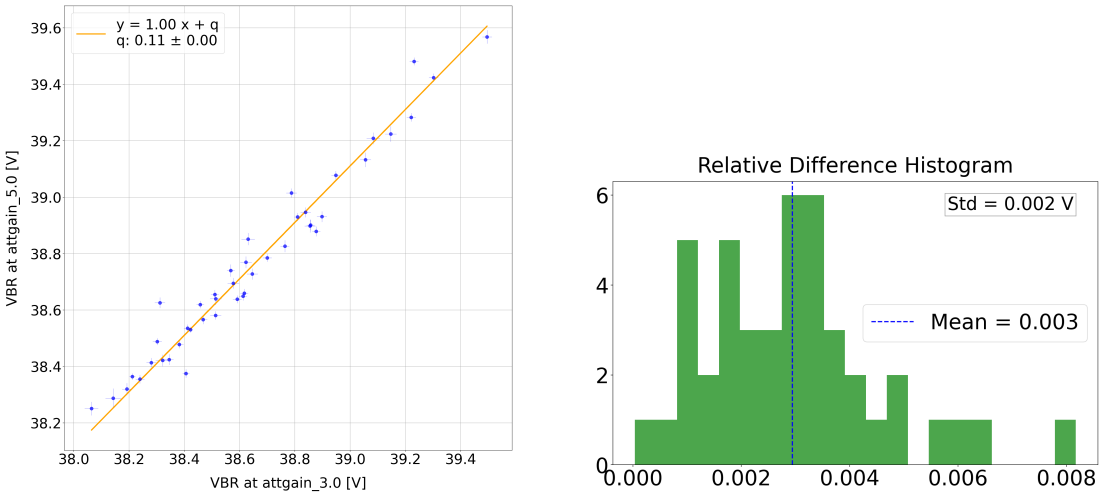


Figure 3.8: Left: Correlation map between the breakdown voltage (VB) estimated from the OV scan using two different values of the attenuation parameter. A linear fit was performed. Right: Relative Difference Histogram between two different VB estimations. The mean and standard deviation are reported. The mean of 0.003 indicates a high level of agreement between the two setups, with minimal spread relative to the associated uncertainties.

A reproducibility test was conducted (fig. 3.8) to verify that the estimation of the VB_{shift} is independent of both the chosen configuration and the specific IV scan run. Additionally, a temperature study was performed by sampling the final values from each temperature sensor on the Readout Unit (RU) across all overvoltage (OV) scans. Each data point corresponds to a temperature measurement from a single sensor during a specific run. Multiple runs were conducted under varying configurations, including different attenuation parameters and OV values. As shown in fig. 3.9, the results demonstrate minimal temperature variation across

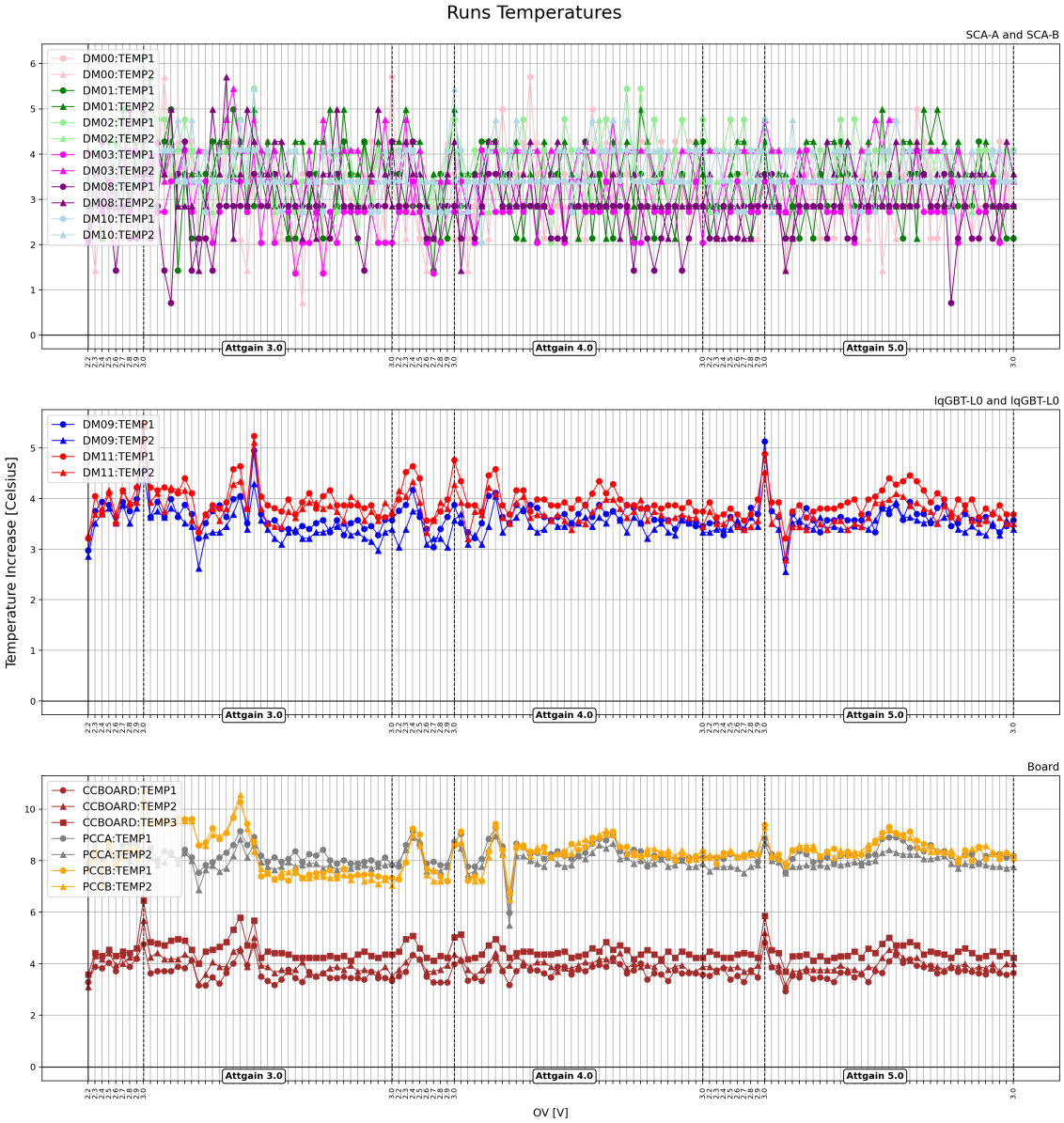


Figure 3.9: Final temperature recorded by each temperature sensor on the Readout Unit (RU) across all over-voltage scans. Each point corresponds to a measurement from a single sensor during a specific run. Multiple runs were performed under different attenuation parameters and OV values. The plot highlights minimal temperature variation between sensors and configurations, confirming stable thermal behaviour throughout the acquisition period.

sensors and runs, confirming that the temperature remains stable and independent of the test conditions throughout the acquisition period.

3.4 Results

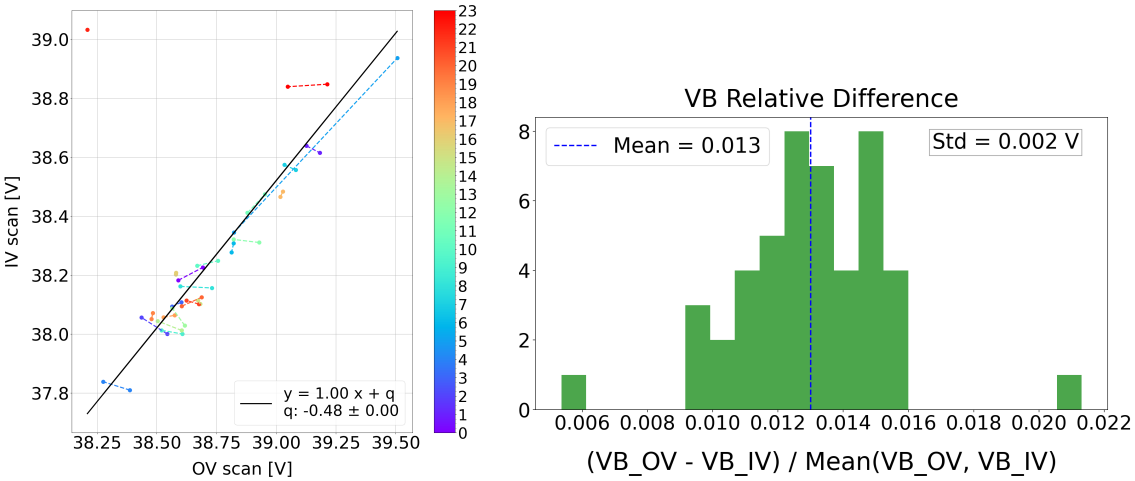


Figure 3.10: Left: Correlation map between the breakdown voltage (VB) estimated from the IV scan and that estimated from the OV scan. A fit with fixed slope was performed to estimate the shift due to temperature variations and systematic bias. A color map is used to identify the SiPM arrays. Pairs of points connected by a line correspond to the two arrays belonging to the same Detector Module (DM). Right: Distribution of the relative difference between the breakdown voltage (VB) estimation and the IV measurement. The mean and standard deviation are reported.

A breakdown voltage estimation was performed on the SiPM arrays of a Readout Unit—one of the six that make up a complete tray. Each measurement was conducted once, using a fixed setting.

Observing fig. 3.10 (left), for DM 12, the breakdown voltage estimation for the 23nd SiPM array couple, one is missing, while another exhibits a shift of approximately 1 V (top-left data point). This discrepancy is due to a failure in the peak extraction process from the energy spectrum which led to the loss of usable data for the fit displayed in fig. 3.7. A dedicated sampling and error estimation procedure—such as the one described in fig. 3.6, bottom—would help mitigate this issue by improving the robustness of peak detection.

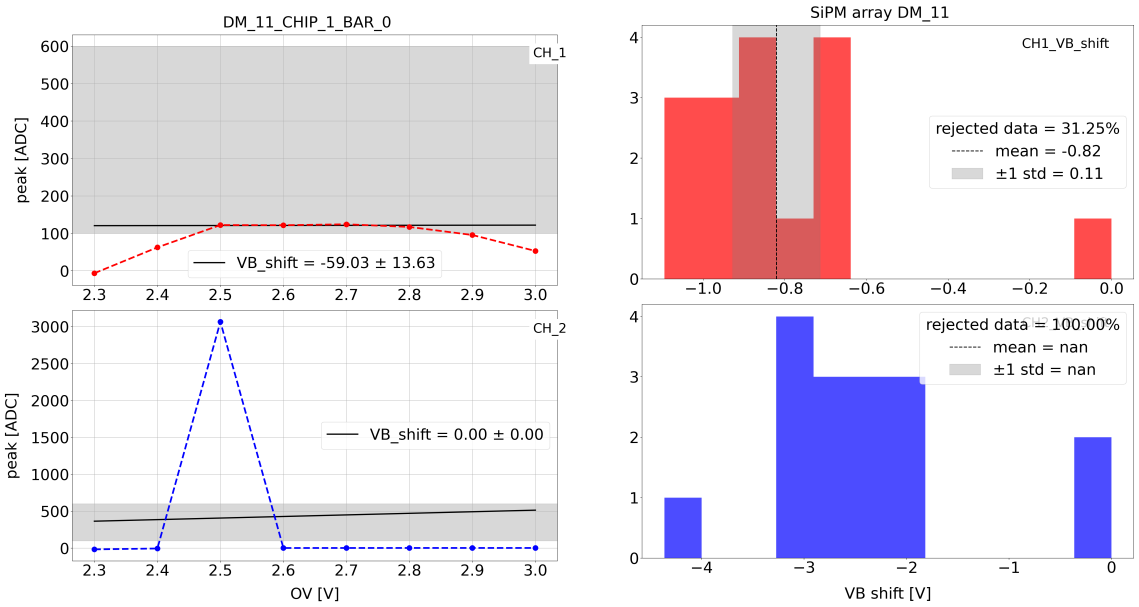


Figure 3.11: Left: Example of a failed peak detection and fit for two SiPMs (one bar), illustrating the difficulties encountered in identifying the energy peak. Right: Histograms of the breakdown voltage (VB) extraction results for the twelve detector modules (DM), showing the distribution and variability across the samples.

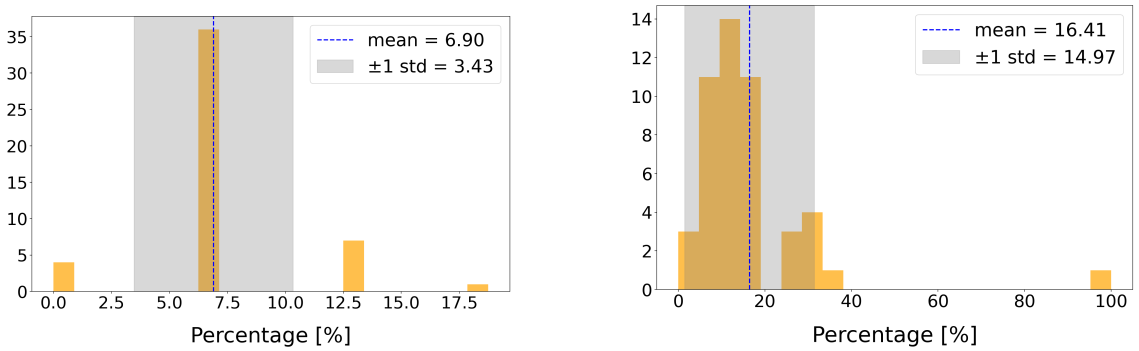


Figure 3.12: Comparison of failure rates in fitting. The left histogram shows the percentage of failed reconstructions (6.90%, std = 3.43) obtained from a larger number of sampled events, leading to better statistics and a lower relative failure rate. In the right histogram, fewer events were sampled, resulting in reduced statistics and a higher failure rate (16.41%, std = 14.97), despite a comparable number of failures. The higher percentage and larger standard deviation reflect the limited statistics rather than a degradation in the performance of the reconstruction.

The problem of unsuccessful peak extraction and fit are clearly illustrated in fig. 3.11, which includes an example of failure in identifying the energy peak, as well as two histograms reporting the rates of failed peak detection and fit reconstruction—with and without applying a statistical study, as shown in fig. 3.12.

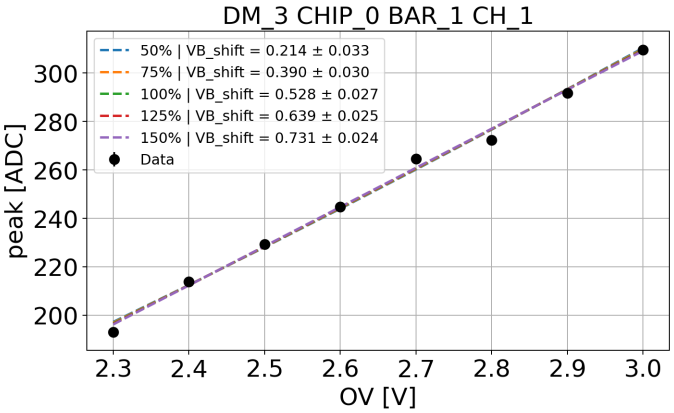
The shift estimated from the fit — approximately 0.493 V with a standard deviation of 0.078 V — between the breakdown voltage (VB) estimated using the over-voltage-based method and the one obtained from IV-curve analysis (see fig. 3.10, left) raises important questions regarding the internal consistency of the method. A temperature difference of about 4 °C can account for roughly 0.148 V of this shift, assuming a typical thermal shift of 37 mV °C^{-1}, but this correction alone is insufficient to fully explain the observed discrepancy. Thus, a remaining shift of approximately 0.345 V persists, indicating that other systematic effects are likely contributing to the offset.

Ideally, if the breakdown voltages derived from the IV measurements are correctly applied in the model and definitions consistently used, then the resulting VB_{shift} should converge to zero. Therefore, testing whether $VB_{\text{shift}} \approx 0$ serves as a closure test — a validation that the $Gain$ and PDE functions, the over-voltage definitions, and the measured spectra are all mutually consistent. Any significant deviation of VB_{shift} from zero would indicate the presence of a systematic bias in the original estimation of the breakdown voltage from IV measurements.

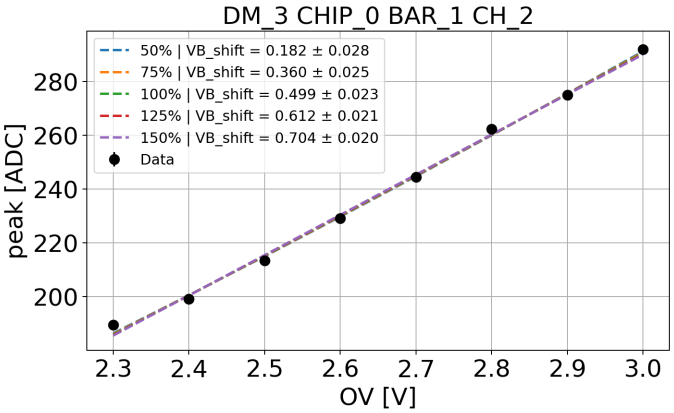
Importantly, this discrepancy does not originate from inconsistent definitions of breakdown voltage across methods. The same IV-based VB values are used as input to for the measurement of both the $Gain$ and PDE curves, so in principle, no shift should appear. However, the persistent offset of about 0.345 V implies a residual systematic effect.

The key insight lies in the nature of the function used to describe the $Gain \times PDE$ behaviour: it is a rigid function, where the only free parameter is the horizontal (voltage) shift. This modelling choice implicitly assumes perfect knowledge (100% accuracy) of all internal parameters — such as slope, offset, and curvature — which are themselves derived from fits performed at non-zero over-voltages, often far from the region near the true breakdown.

This is the central point: because the parameters of the model are extracted from data at finite and relatively high OV values, extrapolating the function toward $OV = 0$ may introduce a non-negligible bias, especially if the original fit is not perfectly accurate. Thus, the observed VB_{shift} could be a natural consequence of unaccounted systematic uncertainties in the model parameters.



(a) Channel 1 of DM_3 CHIP_0 BAR_1



(b) Channel 2 of DM_3 CHIP_0 BAR_1

Figure 3.13: Parameter perturbation study on two channels of a single SiPM bar. The gain-related fit parameters (*Gain* slope and offset) and the *PDE* scaling factor were varied between 50% and 150% of their nominal values in 25% increments (i.e., scaling factors of 0.5, 0.75, 1.0, 1.25, and 1.5). All fits converged successfully, confirming the robustness of the fitting procedure. The resulting breakdown voltage shift, VB_{shift}, ranges from approximately 0.21 V to 0.73 V, consistent with the size of the applied perturbations. For comparison, the residual shift after temperature correction (about 0.345 V) lies within this range. The plots correspond to the data shown in Figure 3.7.

To quantify the sensitivity of the model to such uncertainties, a dedicated study was performed on two channels of a single SiPM bar. The parameters of the $Gain$ and PDE fits (e.g., slope, intercept, and PDE scaling) were systematically perturbed by scaling them from 50% to 150% of their nominal values, in steps of 25%. All fits of the functional form converged successfully, demonstrating that parameter variations within this range do not compromise the overall fitting procedure.

However, the resulting breakdown voltage shift, VB_{shift}, varied significantly — from approximately 0.21 V up to 0.73 V — as parameters were scaled between 50% and 150%. This wide range of shifts highlights the sensitivity of the model to parameter uncertainties. Considering that the remaining shift after temperature correction is about 0.345 V, these results suggest that plausible parameter systematic errors could account for the observed residual offset.

Figure 3.13 shows the impact of these perturbations on the two channels studied, confirming the model's sensitivity and reinforcing the importance of accurately determining fit parameters for reliable extrapolation to zero over-voltage.

It should also be noted that the $Gain \times PDE$ product does not fully describe the SiPM response. Other effects — such as the excess noise factor, optical crosstalk, or saturation effects — may alter the shape of the spectrum and indirectly contribute to the systematic shift.

In conclusion, even under the assumption of coherent definitions and internally consistent procedures, the observed systematic shift in the breakdown voltage cannot be ignored. It likely originates from parameter uncertainties in a rigid model being extrapolated toward $OV = 0$, a region not directly constrained by the data used to build the function. This highlights the importance of critically assessing the extrapolation behaviour of the functional model and recognizing the limits of its predictive accuracy near the breakdown point.

Nevertheless, since the two approaches are based on different measurement principles, exact numerical agreement is not necessarily expected. What matters most is the clear correlation observed between the two estimates. The compatibility analysis, shown in fig. 3.10 (right), compares the two data sets by evaluating the relative difference defined as:

$$\delta = \frac{VB_{OV} - VB_{IV}}{\frac{1}{2}(VB_{OV} + VB_{IV})},$$

which yielded a mean value of 0.013 and a standard deviation of 0.002. This indicates that the relative differences between the breakdown voltage measurements obtained from the OV and IV scans are very small—on average about 1.3% of the mean breakdown voltage—with a limited spread around this value.

In particular, the small standard deviation confirms that the relative residuals are tightly clustered around the expected value. This behaviour is consistent with a high degree of compatibility between the data sets, confirming that the breakdown voltage estimations from both scans agree very well in relative terms.

The results of this study demonstrate a strong correlation and excellent agreement between the OV- and IV-based methods for breakdown voltage estimation. The observed systematic offset is well understood and remains within acceptable limits for standard SiPM operation. These findings confirm the suitability of the OV-scan technique as a reliable and accurate alternative for quality assurance and quality control (QA/QC) procedures in SiPM characterization. Furthermore, the validity of the measurement methods and the quality of the data acquisition and calibration procedures reinforce the robustness of the results presented in this work.

Conclusion

The High-Luminosity Large Hadron Collider (HL-LHC), expected to start operations in 2030, will significantly increase the instantaneous luminosity of proton-proton collisions, posing new challenges due to high pile-up. To mitigate these effects, the CMS experiment will be upgraded with the MIP Timing Detector (MTD), designed to achieve a time resolution of 30–40 ps and enable four-dimensional vertex reconstruction.

The Barrel Timing Layer (BTL), one of the two components of the MTD, is currently under construction at the University of Milano-Bicocca. Its basic detection units consist of LYSO:Ce crystal bars coupled to Silicon Photomultipliers (SiPMs). Each sensor module undergoes dedicated Quality Assurance and Quality Control (QA/QC) procedures after assembly to ensure it meets the required performance specifications.

As part of the international effort to build the Barrel Timing Layer (BTL), I contributed to the production and characterization of the first sensor trays assembled at the Milano-Bicocca assembly center. These early phases were crucial for validating the procedures and infrastructure designed for large-scale QA/QC operations.

Within this framework, I focused specifically on the characterization of the Silicon Photomultipliers (SiPMs) used in the modules. In particular, I worked on methods to evaluate their performance through the analysis of the energy spectrum induced by the intrinsic radioactivity of the LYSO:Ce crystals, aiming to extract key parameters such as the breakdown voltage and to assess the stability of the response under different operating conditions.

An extensive study of the temperature dependence of SiPMs was carried out, providing a crucial parameter to be monitored during the configuration phase of the assembly process.

A new method for determining the breakdown voltage of SiPMs was developed. This approach leverages the intrinsic radioactivity of lutetium in the LYSO crystals—specifically, its β^- decay followed by γ emission—to evaluate energy response behaviour as a function of the over-voltage (OV) applied to the SiPMs. This

technique eliminates the need for an external light source, streamlining the procedure and currently represents the only method available to characterize the SiPMs right after the BTL insertion in CMS. The method has proven successful and has demonstrated its potential for future QA/QC procedures during the on-site assembly phase of the CMS upgrade.

The comparison with the standard IV-based BV estimation shows a strong correlation between the two methods. Quantitatively, the relative difference between the BV values extracted with the two approaches has a mean of 0.013 (1.3%) and the average absolute shift between the breakdown voltages estimated from the OV scan and the IV measurement is approximately 0.493 V.

Despite the encouraging results obtained, the study presents some limitations that suggest directions for future improvements. In particular, the analysis was performed at a single over-voltage value and with limited statistics. Extending the peak extraction and error estimation procedure to cover the full OV scan, supported by improved data selection and a larger dataset, would allow for more accurate and reliable characterization of the detector response. Moreover, implementing a more robust algorithm—capable of identifying peaks even in flat or low-contrast spectra—would reduce dependence on the existing method. A finer analysis at the level of individual SiPM channels could also provide useful insight into the performance of each module. Finally, future work should include a dedicated study on the optimal operating configuration for the final detector. The temperature tests should be repeated using the full QA/QC setup, under passive conditions, to assess heat distribution and thermal uniformity during on-site assembly.

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